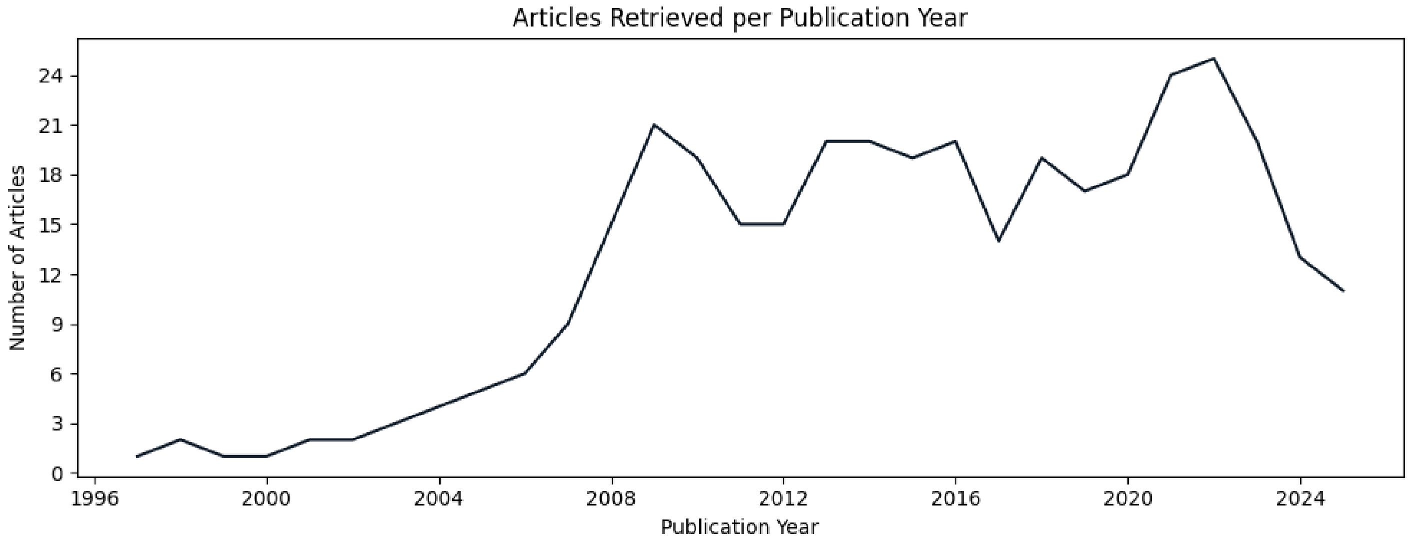
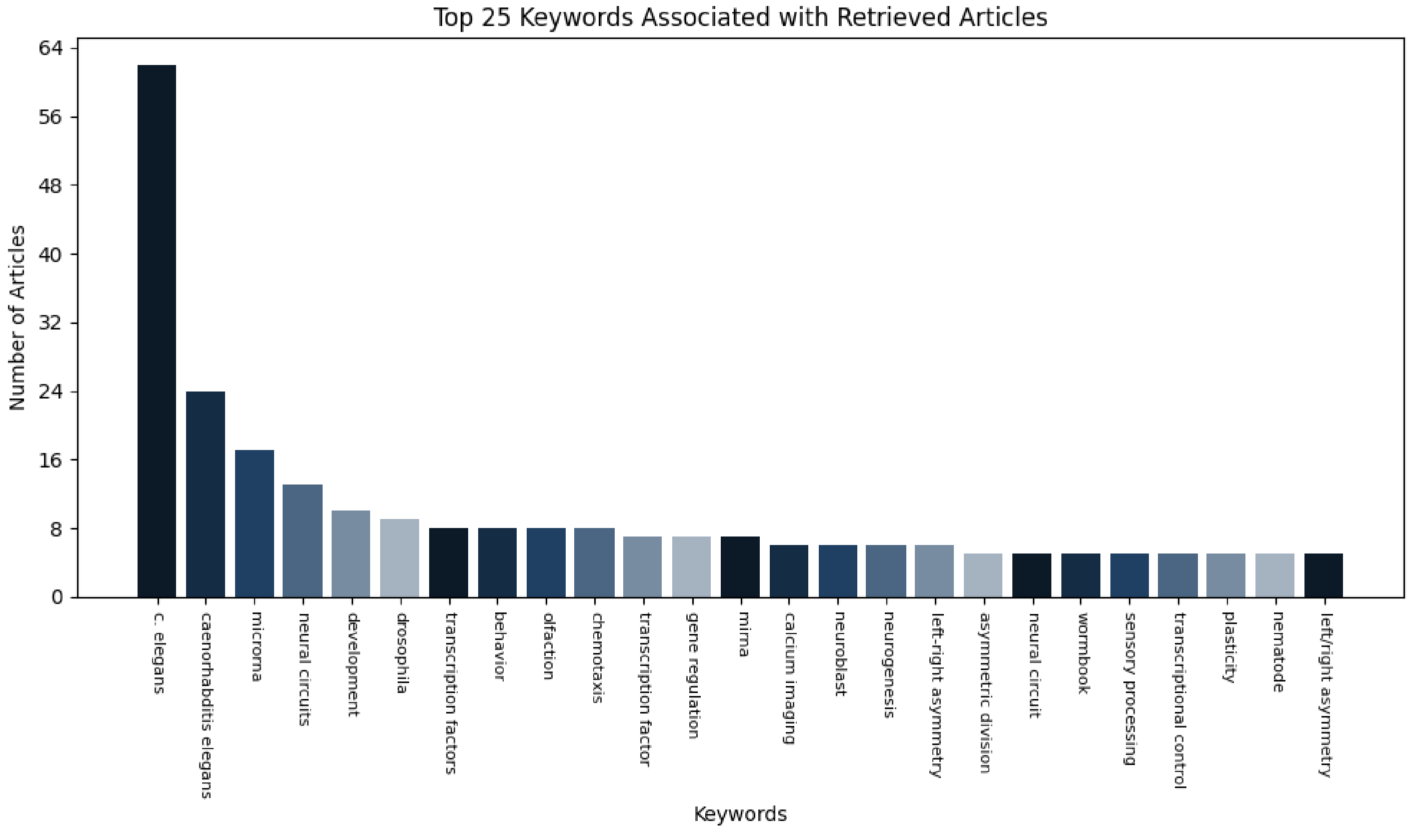
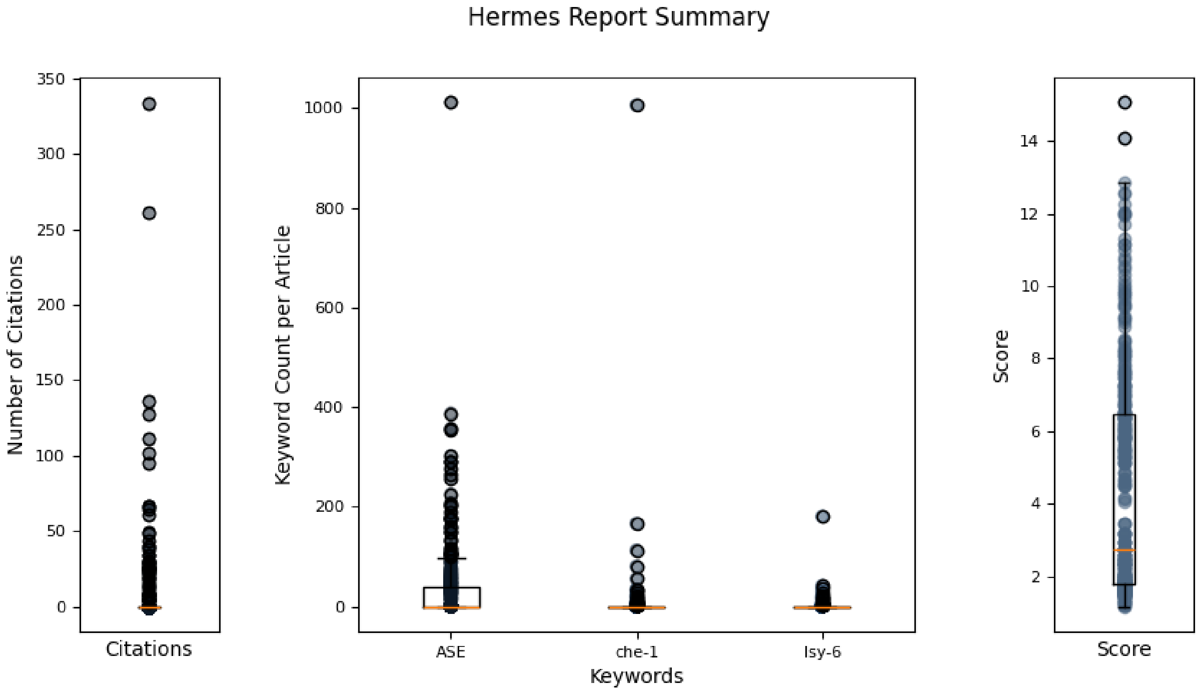
Literature Mining Report

Query Statistics

Query date: 2025-08-25
Query terms: "ASE asymmetry c elegans"
Articles parsed/hits: 364/364 (100.00%)
Scoring keywords: ASE, che-1, lsy-6 (Default Mode)







Query Results

PMID	Year	Journal	Authors	Title	Citations	Score
33693646	2021	Genetics	Ferkey, Denise et al.	chemosensory signal transduction in caenorhab...	66	15.06
33002421	2020	Developmental C	Charest, Julien et al.	combinatorial action of temporally segregated...	27	14.08
25569839	2015	Autophagy	Zhang, Hong et al.	guidelines for monitoring autophagy in caenor...	95	12.85
31526477	2019	eLife	Hong, Ray L et al.	evolution of neuronal anatomy and circuitry i...	34	12.57
24668170	2014	eLife	Doroquez, David et al.	a high-resolution morphological and ultrastru...	127	12.53
39167071	2024	Genetics	Poole, Richard et al.	Neurogenesis in Caenorhabditis elegans	8	12.25
37140557	2023	eLife	Pritz, Christia et al.	principles for coding associative memories in...	13	12.01
19875417	2010	Nucleic Acids R	Takayama, Jun et al.	single-cell transcriptional analysis of taste...	19	12.00
36857454	2023	Science Advance	Lin, Albert et al.	functional imaging and quantification of mult...	25	11.96
32367802	2020	eLife	Schiffer, Jodie et al.	caenorhabditis elegans processes sensory info...	14	11.95
29712787	2018	The Journal of	Bhardwaj, Ashwa et al.	flp-18 functions through the g-protein-couple...	25	11.69
37669262	2023	PLOS Genetics	Tomioka, Masahi et al.	antagonistic regulation of salt and sugar che...	4	11.31
34908528	2021	eLife	Traets, Joleen et al.	mechanism of life-long maintenance of neuron ...	0	11.15
27654945	2016	Molecular Biolo	Lockhead, Dean et al.	the tubulin repertoire of caenorhabditis eleg...	38	11.13
30687001	2019	Frontiers in Mo	Kuramochi, Masa et al.	an excitatory/inhibitory switch from asymmetr...	6	10.97
25032706	2014	PLoS Genetics	Perales, Robert et al.	lin-42, the caenorhabditis elegans period hom...	29	10.78
34237253	2021	Cell	Taylor, Seth R. et al.	molecular topography of an entire nervous sys...	334	10.71
33784383	2021	G3: GenesGenom	Haeussler, Simo et al.	genome-wide rnai screen for regulators of upr...	8	10.55
23874221	2013	PLoS Genetics	Krzyzanowski, M et al.	the c. elegans cgmp-dependent protein kinase ...	22	10.47
37449480	2023	eLife	Brocal-Ruiz, Re et al.	forkhead transcription factor fkh-8 cooperate...	8	10.32
29373666	2018	Chemical Senses	Worthy, Soleil et al.	identification of odor blend used by caenorha...	21	10.16
31488706	2019	Science (New Yo	Packer, Jonatha et al.	a lineage-resolved molecular atlas of c. eleg...	261	10.03
40586702	2025	eLife	NA	evolution of lateralized gustation in nematod...	0	9.96
40224011	2025	iScience	Wang, Ping-Zhou et al.	sensory plasticity caused by up-down regulati...	0	9.87
25738873	2015	PLoS Genetics	Walton, Travis et al.	the bicoid class homeodomain factors ceh-36/o...	25	9.80

Query Result 1/25

Chemosensory signal transduction in Caenorhabditis elegans

Ferkey, Denise M; Sengupta, Piali; LEtoile, Noelle D; Iino, Y

Genetics

Year: 2021 PMID: 33693646 doi: 10.1093/genetics/iyab004 Citations: 66 Score: 15.06

Keywords hits: ASE: 302, che-1: 0, lsy-6: 0

Abstract:

Chemosensory neurons translate perception of external chemical cues, including odorants, tastants, and pheromones, into information that drives attraction or avoidance motor programs. In the laboratory, robust behavioral assays, coupled with powerful genetic, molecular and optical tools, have made Caenorhabditis elegans an ideal experimental system in which to dissect the contributions of individual genes and neurons to ethologically relevant chemosensory behaviors. Here, we review current knowledge of the neurons, signal transduction molecules and regulatory mechanisms that underlie the response of C. elegans to chemicals, including pheromones. The majority of identified molecules and pathways share remarkable homology with sensory mechanisms in other organisms. With the development of new tools and technologies, we anticipate that continued study of chemosensory signal transduction and processing in C. elegans will yield additional new insights into the mechanisms by which this animal is able to detect and discriminate among thousands of chemical cues with a limited sensory neuron repertoire.

Summary:

Thus, calcium signals and depolarization may not always be directly correlated. In addition to their major role in pheromone detection, the ADL neurons play a minor role in chemical avoidance such that their contribution to chemical detection is often revealed only when they are ablated in combination with other sensory neurons. We refer the reader to the primary literature for a more thorough analysis of these gene families and their evolution. GFP-based expression analysis of a subset of the first identified putative receptor genes revealed that many were expressed in only a small subset of chemosensory neurons. The GPA-2 and GPA-3 G proteins have been shown to be necessary for dauer formation, and are expressed in ASK and ASI as well as in other neuron types. Interestingly, consistent with the presence of an octopamine moiety on osas9, avoidance appears to be mediated by the TYRA-2 tyramine/octopamine receptor and the GPA-6 G protein in the ASH nociceptive neurons, although receptors in addition to TYRA-2 expressed in other sensory neuron types may also contribute to the response. Thus, while some Gs may be dedicated to transducing signals from chemosensory GPCRs that detect environmental stimuli, others may also interact with GPCRs that respond to internal signals. Consistent with localization of ODR-3 in the cilia of the AWA, AWB, AWC, ASH, and ADF head sensory neurons, odr-3 mutant animals are highly defective for response to most AWA, AWC, and ASH-detected stimuli, and partly defective for response to 2-nonanone and quinine.

Figures:

Figure 1: Cilia of amphid sensory neurons.

Figure 2: Calcium responses in sensory neurons.

Figure 3: Second messenger levels in cilia versus soma.

Figure 4: Signal transduction pathways in the ASH, AWA, AWC, and ASE sensory neurons.

Figure 5: The RMG hub motor/interneurons are synaptically and electrically connected to O 2 -sensing, nociceptive and pheromone-sensing neurons.

Figure 6: Summarized overview of the chemical structure of ascaroside pheromones.

Figure 7: , bc toggle=yesC.

Mentioned biomedical entities:

Genes: ADF, ADL, ASAP3, ASER, DAF-11, EGL-10, EGL-30, GCs, GOA-1, GPA-12, GPA-13, GPA-14, GPA-2, GPA-3, GPA-5, GPB-1, GPC-1, GPC-2, GSA-1, Gs, ODR-10, ODR-3, PHB, SRG-36, TRP, TYRA-2, gpb-1, gpc-1, srb, unc-31

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron, sperm

Organisms: C. elegans, E. coli, Escherichia coli, Pristionchus pacificus, humans, mice

Tissues: oviduct, pharyngeal muscle

Pathways: N/A

Query Result 2/25

Combinatorial Action of Temporally Segregated Transcription Factors

Charest, Julien; Daniele, Thomas; Wang, Jingkui; Bykov, Aleksandr; Mandlbauer, Ariane; Asparuhova, Mila; Rhsner, Josef; Gutirrez-Prez, Paula; Cochella, Luisa

Developmental Cell

Year: 2020 PMID: 33002421 doi: 10.1016/j.devcel.2020.09.002 Citations: 27 Score: 14.08

Keywords hits: ASE: 179, che-1: 55, lsy-6: 180

Abstract:

Combinatorial action of transcription factors (TFs) with partially overlapping expression is a widespread strategy to generate novel gene-expression patterns and, thus, cellular diversity. Known mechanisms underlying combinatorial activity require co-expression of TFs within the same cell. Here, we describe the mechanism by which two TFs that are never co-expressed generate a new, intersectional expression pattern in C.elegans embryos: lineage-specific priming of a gene by a transiently expressed TF generates a unique intersection with a second TF acting on the same gene four cell divisions later; the second TF is expressed in multiple cells but only activates transcription in those where priming occurred. Early induction of active transcription is necessary and sufficient to establish a competent state, maintained by broadly expressed regulators in the absence of the initial trigger. We uncover additional cells diversified through this mechanism. Our findings define a mechanism for combinatorial TF activity with important implications for generation of cell-type diversity.

Summary:

Early expression of TBX-37/38 is necessary for lsy-6 transcription in ASEL four cell divisions later. These data indicate that the requirement for TBX-37/38 is mediated through a binding site downstream of the lsy-6 miRNA sequence. It was previously shown that early expression of TBX-37/38 could prime a lsy-6 reporter transgene for subsequent activation by CHE-1, but expression of TBX-37/38 after the birth of the ASE neurons was unable to do so. We generated a transgenic strain carrying lineage-specific reporters, synchronized and dissociated embryos, and sorted viable cells from ABa or ABp based on fluorescence, at different time points. The earliest time point at which we could separate cells from the different lineages was at the onset of TBX-37/38 expression in the ABa lineage. We conclude that early adoption of a transcriptionally active state is necessary and sufficient to prime lsy-6 for later robust activation and explains the requirement for TBX-37/38. Given that TBX-37/38 bind the lsy-6 locus early and are not continuously required, we set out to investigate how this competent state of lsy-6 is maintained. Broadly acting TFs could promote low levels of transcription of the lsy-6 locus that, together with the active chromatin marks, may propagate the competent state until the onset of CHE-1. Priming of enhancers prior to the time point when robust transcription is required is a broadly observed phenomenon that can be caused by different mechanisms, e. g., pre-establishment of contact between promoter and distal enhancers, action of pioneer TFs, acquisition of chromatin modifications, or as shown here by promoting early antisense transcription.

Figures:

Figure 1: Integration of Transcriptional Inputs over Time as a Mechanism for Cell Diversification

Figure 2: Early, Direct Binding of TBX-37/38 tolsy-6Is Necessary and Sufficient to Establish Competence for Transcription in the ASE Neurons

Figure 3: TBX-37/38 Are Not Continuously Required forlsy-6Expression

Figure 4: TBX-37/38 Establish a Differentially Accessible State of thelsy-6Locus

Figure 5: Transcription over thelsy-6Locus Occurs Bidirectionally and Is Required for Priming

Figure 6: Asymmetriclsy-6Competence Is Maintained in a SymmetricTrans-acting Factor Environment

Figure 7: TBX-37/38 Determine Left-Right Molecular Asymmetry across Multiple Bilateral Neuron Pairs with ABa/ABp Lineage Origin

Mentioned biomedical entities:

Genes: ASER, B, CHE-1, Cas9, FoxA, L1, PHA-4, TBX-37, TBX-38, TEL, ZIF-1, Zdbf2, galK, gcy-5, histone, lsy-6, unc-119

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, cytosol, neuron, zygote

Organisms: mouse

Tissues: egg, embryo, neural crest

Pathways: N/A

Query Result 3/25

Guidelines for monitoring autophagy in Caenorhabditis elegans

Zhang, Hong; Chang, Jessica T; Guo, Bin; Hansen, Malene; Jia, Kailiang; Kovcs, Attila L; Kumsta, Caroline; Lapierre, Louis R; Legouis, Renaud; Lin, Long; Lu, Qun; Melndez, Alicia; ORourke, Eyleen J; Sato, Ken; Sato, Miyuki; Wang, Xiaochen; Wu, Fan

Autophagy

Year: 2015 PMID: 25569839 doi: 10.1080/15548627.2014.1003478 Citations: 95 Score: 12.85

Keywords hits: ASE: 73, che-1: 0, lsy-6: 5

Abstract:

The cellular recycling process of autophagy has been extensively characterized with standard assays in yeast and mammalian cell lines. In multicellular organisms, numerous external and internal factors differentially affect autophagy activity in specific cell types throughout the stages of organismal ontogeny, adding complexity to the analysis of autophagy in these metazoans. Here we summarize currently available assays for monitoring the autophagic process in the nematode C. elegans. A combination of measuring levels of the lipidated Atg8 ortholog LGG-1, degradation of well-characterized autophagic substrates such as germline P granule components and the SQSTM1/p62 ortholog SQST-1, expression of autophagic genes and electron microscopy analysis of autophagic structures are presently the most informative, yet steady-state, approaches available to assess autophagy levels in C. elegans. We also review how altered autophagy activity affects a variety of biological processes in C. elegans such as L1 survival under starvation conditions, dauer formation, aging, and cell death, as well as neuronal cell specification. Taken together, C. elegans is emerging as a powerful model organism to monitor autophagy while evaluating important physiological roles for autophagy in key developmental events as well as during adulthood.

Summary:

LGG-1 forms punctate structures at the 100-cell stage (and C), but these structures disappear at the comma stage in wild-type embryos (and E). An increased number of LGG-1 puncta and levels of PE-LGG-1 can result from either a block in or induction of autophagic activity and can be detected in mutants with a defect in early steps of autophagosome formation or in a late step of the autophagy pathway such as defective autolysosome function. Elevated autophagy activity during embryogenesis may facilitate the removal of SEPA-1, resulting in a decreased number of SEPA-1 aggregates in early stage mutant embryos compared to wild-type embryos at the same stage. Cautionary notes: SEPA-1 exhibits a dynamic pattern during embryogenesis, so embryos at the same developmental stages should be used for analysis. In loss of autophagy activity mutants, levels of these proteins are dramatically increased and accumulate into aggregates., ZK1053.4 and C35E7.6 colocalize with SQST-1 aggregates, while T04D3.1, T04D3.2 and F44F1.6 aggregates are distinct from SQST-1 aggregates in autophagy mutants. Thus, additional assays, such as GFP:: LGG-1 puncta and levels of LGG-1-II, should be analyzed to examine autophagy. SQST-1:: GFP aggregates in epg-7 mutant larvae and also W07G4.5 aggregates in wild-type larvae are removed by autophagy induced by a variety of stresses.

Figures:

Figure 1: The hierarchical order of autophagy genes in the aggrephagy pathway.

Mentioned biomedical entities:

Genes: ARC, ATG-4.2, ATG-7, ATG16, ATG8, Atg1, Atg10, Atg12, Atg13, Atg16, Atg3, Atg5, Atg7, Atg8, B, C35E7.6, ER, HLH-30, INS, L1, LC3, LGG-1, LGG-2, MOs, P4, PGL-1, PGL-3, RAS, SEPA-1, SQST-1, Vps34, WNT, act-1, ama-1, atg-18, atg-5, atg-7, cog-1, cyn-1, hgrs-1, lgg-1, lsy-6, nhr-23, pmp-3, unc-51

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron, sperm

Organisms: C. elegans, human

Tissues: L4, cuticle, epidermis, hypodermis, oviduct

Pathways: N/A

Query Result 4/25

Evolution of neuronal anatomy and circuitry in two highly divergent nematode species

Hong, Ray L; Riebesell, Metta; Bumbarger, Daniel J; Cook, Steven J; Carstensen, Heather R; Sarpolaki, Tahmineh; Cochella, Luisa; Castrejon, Jessica; Moreno, Eduardo; Sieriebriennikov, Bogdan; Hobert, Oliver; Sommer, Ralf J; Sengupta, Piali; Marder, Eve; Sengupta, Piali

eLife

Year: 2019 PMID: 31526477 doi: 10.7554/eLife.47155.053 Citations: 34 Score: 12.57

Keywords hits: ASE: 90, che-1: 33, lsy-6: 4

Abstract:
The nematodes C. elegans and P. pacificus populate diverse habitats and display distinct patterns of behavior. To understand how their nervous systems have diverged, we undertook a detailed examination of the neuroanatomy of the chemosensory system of P. pacificus. Using independent features such as cell body position, axon projections and lipophilic dye uptake, we have assigned homologies between the amphid neurons, their first-layer interneurons, and several internal receptor neurons of P. pacificus and C. elegans. We found that neuronal number and soma position are highly conserved. However, the morphological elaborations of several amphid cilia are different between them, most notably in the absence of winged cilia morphology in P. pacificus. We established a synaptic wiring diagram of amphid sensory neurons and amphid interneurons in P. pacificus and found striking patterns of conservation and divergence in connectivity relative to C. elegans, but very little changes in relative neighborhood of neuronal processes. These findings demonstrate the existence of several constraints in patterning the nervous system and suggest that major substrates for evolutionary novelty lie in the alterations of dendritic structures and synaptic connectivity.

Summary:
All amphid neurons and the amphid sheath cell in P. pacificus have their cell bodies in the lateral ganglion posterior to the nerve ring, which resembles the condition found in C. elegans and other nematodes. Similar to AFD endings in other species, the AM12 cilium is fully embedded in the amphid sheath cell process, does not enter the lumen of the sheath cell or the amphid channel and thus has no contact to the outside environment. As the AM7 and AWC axons both cross the dorsal midline and terminate above the lateral midline, we regard them as homologs based on this singular feature, although the AM7 cell body appears to have shifted from a position ventral of ASH in C. elegans to a position between AM1 and AM8 in P. pacificus, such that the three cell types are just ventral and parallel to the lateral midline on each side. Two other amphid neurons also show strong resemblance to their C. elegans counterparts according to their unique axon projections. Four of the strong gap junction connections were present in both species while 6 and 2 were specific to P. pacificus and C. elegans, respectively. Hence, like odr-7 expression, the conservation of cell-type specific expression is limited to the amphid neurons but not to the proposed cellular homologs when soma position and axon projection patterns are also considered. In the absence of morphological characteristics such as the winged endings of AWx neurons in C. elegans, we proposed neuronal homologies between P. pacificus and C. elegans amphid neurons based on their relative cell body positions, axon projections, the manner of dendrite entry into the amphid sheath cell, the number of cilia in channel, dye filling properties, and the connections to first layer interneurons.

Figures:

Figure 1: EM of the amphid sensillum and two other sensory neurons ofP. pacificushermaphrodite adults.

Figure 2: URX is ciliated.

Figure 3: URY, URX, URA and BAG endings.

Figure 4: A comparison of select non-amphid ciliated neurons in three nematode species.

Figure 5: Video of URX 3D reconstruction.

Figure 6: Video of URY 3D reconstruction.

Figure 7: Video of URA 3D reconstruction.

Figure 8: Position of amphid cilia in AMso and AMsh.

Figure 9: Posterior-anterior path of amphid dendrite entry into the left sheath glia of specimen 107.

Figure 10: Posterior-anterior path of amphid dendrite entry into the right sheath glia of specimen 107.

Figure 11: Overview of the amphid sensilla.

Figure 12: 3D rendering of AM1(ASH) neurons.

Figure 13: 3D rendering of AM2(ADL) neurons.

Figure 14: 3D rendering of AM3(AWA) neurons.

Mentioned biomedical entities:

Genes: ADF, AM5, ASER, B, BAG, FLP, PHB, T12, URB, gcy-6, lsy-6

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, cell body, cilium, epidermal cell, neuron, secretory cell, sheath cell

Organisms: A. complexus, Acrobeles complexus, C. elegans, E. coli, H. contortus, Haemonchus contortus, Parastrongyloides trichosuri, swine

Tissues: anterior, cuticle, dorsal, dorsal nerve cord, excretory duct

Pathways: N/A

Query Result 5/25

A high-resolution morphological and ultrastructural map of anterior sensory cilia and glia in Caenorhabditis elegans

Doroquez, David B; Berciu, Cristina; Anderson, James R; Sengupta, Piali; Nicastro, Daniela; Hobert, Oliver; Hobert, Oliver

eLife

Year: 2014 PMID: 24668170 doi: 10.7554/eLife.01948.039 Citations: 127 Score: 12.53

Keywords hits: ASE: 49, che-1: 0, lsy-6: 0

Abstract:

Many primary sensory cilia exhibit unique architectures that are critical for transduction of specific sensory stimuli. Although basic ciliogenic mechanisms are well described, how complex ciliary structures are generated remains unclear. Seminal work performed several decades ago provided an initial but incomplete description of diverse sensory cilia morphologies in C. elegans. To begin to explore the mechanisms that generate these remarkably complex structures, we have taken advantage of advances in electron microscopy and tomography, and reconstructed three-dimensional structures of fifty of sixty sensory cilia in the C. elegans adult hermaphrodite at high resolution. We characterize novel axonemal microtubule organization patterns, clarify structural features at the ciliary base, describe new aspects of ciliaglia interactions, and identify structures suggesting novel mechanisms of ciliary protein trafficking. This complete ultrastructural description of diverse cilia in C. elegans provides the foundation for investigations into underlying ciliogenic pathways, as well as contributions of defined ciliary structures to specific neuronal functions.

Summary:

In particular, these studies described the morphologically diverse cilia present at the dendritic endings of sensory neuron types in these sensilla. For instance, the morphology and ultrastructure of cilia in the BAG gas-sensing and FLP nociceptive neurons are poorly described, and the axonemal MT distribution pattern in highly branched cilia, such as those in the AWA chemosensory neurons, have not been fully characterized. Here, we provide a high-resolution morphological and ultrastructural analysis of the anterior endings of sensory neurons and glia in the adult C. elegans hermaphrodite using ssTEM and ssET of high-pressure frozen and freeze-substituted adult animals. This matrix-filled space is especially pronounced in the amphid organs, where the amphid sheath cell process surrounds a large lumen that tapers into a narrower channel distally that is open to the environment. The prototypical cilium in adult C. elegans can be separated into four major segments: a distal segment, a middle segment that comprises the main axonemal core consisting of nine outer doublet MTs, a TZ at the base of the axoneme, and a bulb-like periciliary membrane compartment proximal to the TZ. Each ciliary rod contains the full complement of MT features found in single-rod channel cilia, including nine sMTs in their distal segments and nine dMTs in their middle segments along with 46 isMTs that originate at the TZ region. The proximal origin of the nine dMTs of these cilia, and MT organization at their TZs has not been previously described. The nine Y-linked dMTs at the TZs of each ciliary rod of ADL and ADF are found within a tight ciliary shaft that flares as the ciliary base widens, so that the proximal cross-sections often show an oval array of 18 dMTs. Each AWB neuron has two ciliary branches with variable lengths and morphologies.

Figures:

Figure 1: TEM cross-section with identified ciliary and glial endings.

Figure 2: Ultrastructural features maintained by HPF-FS.

Figure 3: Example TEM cross-section images of theC. elegansnose.

Figure 4: Graphical model of a high-resolution ssTEM 3D reconstruction of the anterior sensory endings.

Figure 5: Two glial cell types are associated with each ciliated sensillum.

Figure 6: 3D reconstruction models of glia associated with different sensilla.

Figure 7: CeAJ connections between glial processes and surrounding cells.

Figure 8: ssTEM 3D reconstruction model of all ciliated and non-ciliated anterior sensory endings in theC. elegansadult hermaphrodite. Color codes are as indicated inFigure 2.

Figure 9: C. elegansciliary ultrastructure.

Figure 10: Subcellular features of amphid cilia.

Figure 11: ET analyses of flared dMTs at the ciliary base.

Figure 12: Vesicles at the ciliary base and in the TZ.

Figure 13: The amphid sensillum.

Figure 14: ssTEM 3D reconstruction model of all amphid neuron cilia and associated socket and sheath cell processes. Color codes as indicated inFigure 2.

Figure 15: Morphology and ultrastructure of amphid channel cilia.

Figure 16: 3D reconstruction model of axonemal MTs twisting as the ASI cilium projects distally. Dark purpleA-tubules; light purpleB-tubules.

Figure 17: Ultrastructure of the ADL neuron cilia.

Figure 18: Ultrastructure of the ADF neuron cilia.

Figure 19: 3D reconstruction model of MT distribution at the TZs of the two cilia in ADL amphid neurons. Color codes for MTs as indicated inFigure 8legend.

Figure 20: Ultrastructure of the AWB neuron cilia.

Figure 21: Ultrastructure of the AWA cilium.

Figure 22: 3D reconstruction model of MT distribution in the branches of the AWA cilium. Color codes for MTs as indicated inFigure 10legend.

Figure 23: Ultrastructure of the AWC cilium.

Figure 24: Ultrastructure of the sensory endings of AFD.

Figure 25: Periciliary membrane compartments.

Figure 26: Interactions between the PCMCs of AFD and ASE.

Figure 27: Ultrastructure of IL1 and IL2 neurons.

Figure 28: Structural features associated with lateral IL1 and IL2 cilia.

Figure 29: 3D reconstruction model of the sensory endings and associated glial cells processes of the IL, OLL/OLQ and CEP neurons.

Figure 30: Ultrastructure of the OLL/OLQ and CEP neuron cilia.

Figure 31: Structural features associated with OLQ and CEP cilia.

Figure 32: Ultrastructure of BAG and FLP cilia.

Figure 33: 3D reconstruction model of BAG and FLP cilia associated with the ILso process.

Figure 34: Non-ciliated endings of URX and URY.

Mentioned biomedical entities:
Genes: ADF, ADL, BAG, FLP, IL1, IL2, MT, MTs, PA, URB

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, cilium, glial cell, neuron, sheath cell

Organisms: C. elegans

Tissues: cuticle, dorsal, sensillum, tissue, tuft

Pathways: N/A

Query Result 6/25

Neurogenesis in Caenorhabditis elegans

Poole, Richard J; Flames, Nuria; Cochella, Luisa; Sundaram, M

Genetics

Year: 2024 PMID: 39167071 doi: 10.1093/genetics/iyae116 Citations: 8 Score: 12.25

Keywords hits: ASE: 149, che-1: 34, lsy-6: 17

Abstract:

Animals rely on their nervous systems to process sensory inputs, integrate these with internal signals, and produce behavioral outputs. This is enabled by the highly specialized morphologies and functions of neurons. Neuronal cells share multiple structural and physiological features, but they also come in a large diversity of types or classes that give the nervous system its broad range of functions and plasticity. This diversity, first recognized over a century ago, spurred classification efforts based on morphology, function, and molecular criteria. Caenorhabditis elegans, with its precisely mapped nervous system at the anatomical level, an extensive molecular description of most of its neurons, and its genetic amenability, has been a prime model for understanding how neurons develop and diversify at a mechanistic level. Here, we review the gene regulatory mechanisms driving neurogenesis and the diversification of neuron classes and subclasses in C. elegans. We discuss our current understanding of the specification of neuronal progenitors and their differentiation in terms of the transcription factors involved and ensuing changes in gene expression and chromatin landscape. The central theme that has emerged is that the identity of a neuron is defined by modules of gene batteries that are under control of parallel yet interconnected regulatory mechanisms. We focus on how, to achieve these terminal identities, cells integrate information along their developmental lineages. Moreover, we discuss how neurons are diversified postembryonically in a time-, genetic sex-, and activity-dependent manner. Finally, we discuss how the understanding of neuronal development can provide insights into the evolution of neuronal diversity.

Summary:

Together, these approaches revealed that combinations of key TFs are necessary for defining neuron class identity and act, for the most part, by directly activating the expression of broad batteries of terminal effector genes. In the section on neuronal diversification, we go deeper into how combinations of terminal selectors achieve neuron class-specific gene expression and how these combinations are generated during development. Mutational analysis of terminal selectors in multiple different contexts has revealed that while most neuron-specific genes fail to be normally expressed, there are at least 2 gene batteries that are typically unaffected: panneuronal genes, which as explained above are primarily regulated by the CUT TFs, and genes coding for structural cilia components, which are shared by all sensory ciliated neurons. Other key regulators of neural precursor specification in both Drosophila and vertebrates include the HOX cluster HD TFs. As discussed below, many but not all these genes have been shown to regulate neural vs nonneural decisions and the specification of neural progenitors. The discovery in Drosophila that achaetescute controls the development of the external sensory organ precursors and that atonal controls the development of the internal chordotonal organs led to the realization that proneural genes not only regulate the specification of neuronal precursors but may also play important roles in the differentiation of different classes of neurons. This classic proneural function is not restricted to postembryonic lineages as lin-32 is also expressed and required in the ABplaaaaa/ABarpapaa neuroblasts that give rise respectively to the left and right URAD, CEPD, and URX neurons. Several other C. elegans proneural genes have also been shown to regulate the specification of neuroblasts including the AS-C homolog hlh-14, which is required for the embryonic specification of the PVQ/HSN/PHB neuroblasts, the ABalpppp/ABpraaap neuroblasts, the DVC neuroblast, and the PVR neuron the NeuroD homolog cnd-1, which is required for the specification of the neuroblasts that give rise to the embryonic DA, DB, and DD ventral nerve cord motor neurons and the Neurogenin homolog ngn-1, which is required for the LR asymmetric specification of the MI neuron.

Figures:

Figure 1: Regulatory framework for C. elegans neurogenesis.

Figure 2: Developmental origin of neurons and neuronal diversity.

Figure 3: Terminal selector functions in neuronal diversification.

Figure 4: Generation of terminal selector combinations in space and time.

Figure 5: Repressor-based mechanisms for neuronal diversification.

Figure 6: Environmental effects on neuron diversity.

Figure 7: Progressive determination model of neuronal development.

Figure 8: Expression of bHLH TFs in neuronal lineages.

Mentioned biomedical entities:

Genes: ASE, AST-1, B, CEH-10, CHE-1, CUT, DAF-19, DDs, HD, Hh, IL2, MEC-3, MyoD, NGF, Neurogenin, Notch, ONECUT, PHB, RIP, Sox, Sox-3, Sox2, SoxB, SoxC, UNC-18, UNC-3, UNC-86, ceh-13, ces-1, ces-2, egl-5, hlh-3, hlh-6, lin-32, lin-39, mab-5, ngn-1, nob-1, php-3, sox-2, unc-3, unc-30, unc-4

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, cilium, ectoderm, glial cell, mesoderm, muscle cell, neural progenitor cell, neural stem cell, neuron

Organisms: C. elegans, humans, mice, zebrafish

Tissues: hypodermis, nervous system, neural tissue, tissue

Pathways: N/A

Query Result 7/25

Principles for coding associative memories in a compact neural network

Pritz, Christian; Itskovits, Eyal; Bokman, Eduard; Ruach, Rotem; Gritsenko, Vladimir; Nelken, Tal; Menasherof, Mai; Azulay, Aharon; Zaslaver, Alon; Iino, Yuichi; Behrens, Timothy E; Iino, Yuichi; Iino, Yuichi

eLife

Year: 2023 PMID: 37140557 doi: 10.7554/eLife.74434.sa2 Citations: 13 Score: 12.01

Keywords hits: ASE: 130, che-1: 0, lsy-6: 0

Abstract:

A major goal in neuroscience is to elucidate the principles by which memories are stored in a neural network. Here, we have systematically studied how four types of associative memories (short- and long-term memories, each as positive and negative associations) are encoded within the compact neural network of Caenorhabditis elegans worms. Interestingly, sensory neurons were primarily involved in coding short-term, but not long-term, memories, and individual sensory neurons could be assigned to coding either the conditioned stimulus or the experience valence (or both). Moreover, when considering the collective activity of the sensory neurons, the specific training experiences could be decoded. Interneurons integrated the modulated sensory inputs and a simple linear combination model identified the experience-specific modulated communication routes. The widely distributed memory suggests that integrated network plasticity, rather than changes to individual neurons, underlies the fine behavioral plasticity. This comprehensive study reveals basic memory-coding principles and highlights the central roles of sensory neurons in memory formation.

Summary:

In parallel to the trained animals, we always included in the analysis two important control groups: mock-trained animals and naive animals, which were left untreated. To verify that these training paradigms form robust memory traces, we analyzed the attraction of trained animals to BUT, the CS. Most striking, the changes in neural responses were observed predominantly for the short-term training paradigms, while only minor negligible changes were detected following long-term training paradigms. In contrast to sensory neurons, interneurons showed marked changes in their responses following training in both short- and long-term paradigms. For this, we averaged activities of each odor trial for each of the sensory neurons across all animals in the different paradigms, and used a multivariate regression analysis to extract the weights that would best fit the activity of the AIY neurons. For this, we considered the changes in the activity dynamics of individual neurons across the different paradigms by calculating the difference in the neural activity between the training groups. Notably, since this PCA-based method filters much of the variation in the data that is not associated with the experience components of the training paradigms, it provides a conservative, and presumably more accurate, representation for the role of neurons in each of the training paradigms. It is evident that coding each of the memory components is distributed among several sensory neurons, where a few of them are broadly used to code most of these experience components.

Figures:

Figure 1: Training paradigms that form robust associative memories.

Figure 2: Behavioral choice assays using ethanol as the alternative choice reproduced the results obtained using diacetyl.

Figure 3: Appetitive and aversive training paradigms shift the preference to butanone.

Figure 4: Inferring memory components by comparing different experimental groups.

Figure 5: The behavioral choice assays are subject to high day-to-day variability and a same-day difference analysis reduced this variation.

Figure 6: A comprehensive functional analysis of the chemosensory system including key interneurons.

Figure 7: Identifying the individual chemosensory neurons in the pan-chemosensory reporter strain.

Figure 8: Interneurons anatomy and the region of extracted activity.

Figure 9: Discriminating between the two bilateral AWC neurons.

Figure 10: Discriminating between the URX and CEPD neurons: Butanone removal elicits responses in the URX neurons, but not in the CEPD neurons.

Figure 11: Neural activities of naive animals.

Figure 12: Individual activity traces of chemosensory neurons in naive and mock-controlled animals.

Figure 13: Interneurons exhibited modulated activities following both short- and long-term training paradigms.

Figure 14: Individual activity traces of interneurons in naive and mock-controlled animals.

Figure 15: Short-term training broadly modulates activity of the sensory neurons.

Figure 16: A comprehensive functional analysis of the chemosensory system and key interneurons following training in short-term paradigms.

Figure 17: Chemosensory neurons showing modulated activities following short-term training paradigms.

Figure 18: Individual activity traces of chemosensory neurons in naive and mock-controlled animals.

Figure 19: The two AWA reporter lines differ in their response kinetics but carry the same memory-coding logic.

Figure 20: The WT and the pan-chemosensory reporter strain show similar behavioral outputs following training.

Figure 21: ASER and AWA neurons show high animal-to-animal variability.

Figure 22: Activity of sensory neurons can be used to identify the training conditions by classification algorithms.

Figure 23: Training-induced modulated activity of the interneurons.

Figure 24: Stimulus-induced calcium dynamics is observed in the neurites of AIA and AIY, but not in the cell soma.

Figure 25: The individual activity traces and the mean activities of RIA and AIY interneurons in all training paradigms.

Figure 26: Activities in the AVA and AVE neurons were not modulated following training in the different paradigms.

Figure 27: Activity of AIY interneurons can be described as a linear combination of the sensory neurons activities.

Figure 28: Synaptic communication routes between sensory neurons and the AIY interneurons change in an experience-dependent manner.

Figure 29: PC analysis reveals unique population codes for each of the training paradigms.

Figure 30: PC analysis reveals the encoding neurons in each of the training paradigms.

Figure 31: Short-term learning paradigms modulated the directionality towards the CS, but not the speed nor the reversal frequency.

Figure 32: Simulations of choice behavior based on measured locomotion parameters.

Figure 33: Illustration of the sensory- and the inter- neurons participating in coding the different memory types.

Figure 34: The activity of sensory neurons is predominantly modulated following short-term training paradigms, while the activity of interneurons is modulated following both short- and long-term training paradigms.

Figure 35: The gained responses in AWCONfollowing short-term appetitive training require intact synaptic transmission.

Figure 36: The sensory neurons whose activity responses were modulated following short-term, but not long-term, training paradigms.

Mentioned biomedical entities:

Genes: ADF, CS, IX, P40

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: C. elegans

Tissues: nervous system, tube

Pathways: N/A

Query Result 8/25

Single-cell transcriptional analysis of taste sensory neuron pair in Caenorhabditis elegans

Takayama, Jun; Faumont, Serge; Kunitomo, Hirofumi; Lockery, Shawn R.; Iino, Yuichi

Nucleic Acids Research

Year: 2010 PMID: 19875417 doi: 10.1093/nar/gkp868 Citations: 19 Score: 12.00

Keywords hits: ASE: 263, che-1: 17, lsy-6: 9

Abstract:

The nervous system is composed of a wide variety of neurons. A description of the transcriptional profiles of each neuron would yield enormous information about the molecular mechanisms that define morphological or functional characteristics. Here we show that RNA isolation from single neurons is feasible by using an optimized mRNA tagging method. This method extracts transcripts in the target cells by co-immunoprecipitation of the complexes of RNA and epitope-tagged poly(A) binding protein expressed specifically in the cells. With this method and genome-wide microarray, we compared the transcriptional profiles of two functionally different neurons in the main C. elegans gustatory neuron class ASE. Eight of the 13 known subtype-specific genes were successfully detected. Additionally, we identified nine novel genes including a receptor guanylyl cyclase, secreted proteins, a TRPC channel and uncharacterized genes conserved among nematodes, suggesting the two neurons are substantially different than previously thought. The expression of these novel genes was controlled by the previously known regulatory network for subtype differentiation. We also describe unique motif organization within individual gene groups classified by the expression patterns in ASE. Our study paves the way to the complete catalog of the expression profiles of individual C. elegans neurons.

Summary:

The primers used for the amplification of each gene were: lmn152: 5-CGTTCACCACCCACCAGAA-3, lmn132: 5-CAAGACGAGCTGATGGGTTATCT-3 for lmn-1 gcy551: 5-CCTACCAAGAGAAAAAGTTGAACTAAGAA-3, gcy531: 5-GGGCATGGATAGACCAACGA-3 for gcy-5 gcy651: 5-CACGTGGCGAAGTCATAATCA-3, gcy6-31: 5-GCTGACTCGTCCATTTTAAGCA-3 for gcy-6 gcy7. fw1: 5-TCTCCCAGACCCGATTTGG-3, gcy7. rv1: 5-CCGAAGGACGTTCGGTAAAAC-3 for gcy-7. Most of the genes expressed in ASEL and/or ASER have ASE motifs in their 5-upstream regions, which are specifically required for the expression in ASE neurons. We found that all of the novel ASER-biased genes had ASE motif in the promoter regions, whereas no ASE motif was found in the ASEL-biased ins-32 promoter. Therefore, we suggest that all of the eight novel ASER-biased genes could be directly regulated by CHE-1 binding to their ASE motifs whereas ins-32 expression is regulated indirectly. To determine whether the ASE motifs were enriched in the genes expressed in ASE neurons, we examined the frequency of ASE motifs per one kilobase promoter of each gene set categorized by the expression patterns in ASE. The frequency was higher in the genes expressed in either or both of ASE neurons than those not expressed in ASE neurons or those selected randomly from the microarray probe set, suggesting that the ASE motif is enriched in the genes expressed in ASE neurons.

Figures:

Figure 1: FLAG::PABP is expressed in single neurons of the transgenic animals.

Mentioned biomedical entities:

Genes: ASER, Bad, C14C6.3, CHE-1, L1, PABP, TRPC, cog-1, die-1, flp, fozi-1, gcy-22, gcy-5, gcy-6, ins-32, lim-6, lsy-6, nlp-5

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: C. elegans

Tissues: C7, hypodermis, medial, nervous system

Pathways: N/A

Query Result 9/25

Functional imaging and quantification of multineuronal olfactory responses in C. elegans

Lin, Albert; Qin, Shanshan; Casademunt, Helena; Wu, Min; Hung, Wesley; Cain, Gregory; Tan, Nicolas Z.; Valenzuela, Raymond; Lesanpezeshki, Leila; Venkatachalam, Vivek; Pehlevan, Cengiz; Zhen, Mei; Samuel, Aravinthan D.T.

Science Advances

Year: 2023 PMID: 36857454 doi: 10.1126/sciadv.ade1249 Citations: 25 Score: 11.96

Keywords hits: ASE: 66, che-1: 0, lsy-6: 0

Abstract:

Many animals perceive odorant molecules by collecting information from ensembles of olfactory neurons, where each neuron uses receptors that are tuned to recognize certain odorant molecules with different binding affinity. Olfactory systems are able, in principle, to detect and discriminate diverse odorants using combinatorial coding strategies. We have combined microfluidics and multineuronal imaging to study the ensemble-level olfactory representations at the sensory periphery of the nematode Caenorhabditis elegans. The collective activity of C. elegans chemosensory neurons reveals high-dimensional representations of olfactory information across a broad space of odorant molecules. We reveal diverse tuning properties and dose-response curves across chemosensory neurons and across odorants. We describe the unique contribution of each sensory neuron to an ensemble-level code for volatile odorants. We show that a natural stimuli, a set of nematode pheromones, are also encoded by the sensory ensemble. The integrated activity of the C. elegans chemosensory neurons contains sufficient information to robustly encode the intensity and identity of diverse chemical stimuli.

Summary:

However, using selected odorants to stimulate single sensory neurons and evoke behavioral responses does not reveal how olfactory inputs might be encoded by the sensory neuron ensemble. We found that most olfactory stimuli activate multiple chemosensory neurons in C. elegans. And A. D. T. S. And V. V. And V. V. And S. Q.

Figures:

Figure 1: Labeling and recording from chemosensory neurons.

Figure 2: Ensemble responses to a broad odorant panel.

Figure 3: Odorant representations in synaptic transmission mutants.

Figure 4: Chemosensory neuron tuning.

Figure 5: Representative comparisons of single-trial odorant responses.

Figure 6: Odorant discriminability is robust to virtual knockouts.

Figure 7: Odorant representations of pheromones.

Mentioned biomedical entities:

Genes: ADF, ASER, B, GPCR, ODR-10, P40, gpc-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: C. elegans, Escherichia coli

Tissues: N/A

Pathways: N/A

Query Result 10/25

Caenorhabditis elegans processes sensory information to choose between freeloading and self-defense strategies

Schiffer, Jodie A; Servello, Francesco A; Heath, William R; Amrit, Francis Raj Gandhi; Stumbur, Stephanie V; Eder, Matthias; Martin, Olivier MF; Johnsen, Sean B; Stanley, Julian A; Tam, Hannah; Brennan, Sarah J; McGowan, Natalie G; Vogelaar, Abigail L; Xu, Yuyan; Serkin, William T; Ghazi, Arjumand; Stroustrup, Nicholas; Apfeld, Javier; Hobert, Oliver; Calabrese, Ronald L; Hobert, Oliver

eLife

Year: 2020 PMID: 32367802 doi: 10.7554/eLife.56186.sa2 Citations: 14 Score: 11.95

Keywords hits: ASE: 175, che-1: 0, lsy-6: 0

Abstract:
Hydrogen peroxide is the preeminent chemical weapon that organisms use for combat. Individual cells rely on conserved defenses to prevent and repair peroxide-induced damage, but whether similar defenses might be coordinated across cells in animals remains poorly understood. Here, we identify a neuronal circuit in the nematode Caenorhabditis elegans that processes information perceived by two sensory neurons to control the induction of hydrogen peroxide defenses in the organism. We found that catalases produced by Escherichia coli, the nematodes food source, can deplete hydrogen peroxide from the local environment and thereby protect the nematodes. In the presence of E. coli, the nematodes neurons signal via TGF-insulin/IGF1 relay to target tissues to repress expression of catalases and other hydrogen peroxide defenses. This adaptive strategy is the first example of a multicellular organism modulating its defenses when it expects to freeload from the protection provided by molecularly orthologous defenses from another species.

Summary:
Because ablation of ASI increased peroxide resistance but ablation of ASJ did not, we reason that ASI neurons are the source of DAF-7 that regulates the nematodes peroxide resistance. We next asked whether DAF-7/TGF from ASI might regulate resistance to additional toxic chemicals from the environment that are not peroxides or directly generate peroxides. We found that DAF-16 was also necessary for the increase in peroxide resistance of daf-2 mutants and for the increase in peroxide resistance of an ins-4 ins-5 ins-6 daf-28 quadruple mutant. We propose that repression of insulin/IGF1 gene expression by DAF-3/coSMAD leads to a reduction in signaling by the DAF-2/insulin/IGF1 receptor, which subsequently increases the nematodes peroxide resistance via transcriptional activation by SKN-1/NRF and DAF-16/FOXO. To identify which target tissues are important for increasing the nematodes peroxide resistance via DAF-16 in response to reduced DAF-1 signaling, we determined the extent to which restoring daf-16 expression in specific tissues using tissue-specific promoters increased the peroxide resistance of daf-16 daf-1 double mutants. We propose that E. coli protects C. elegans from hydrogen peroxide killing because it expresses catalases that efficiently deplete hydrogen peroxide from the environment, creating a local environment where hydrogen peroxide is not a threat to C. elegans. To determine whether DAF-7 regulates C. elegans hydrogen peroxide resistance, similar to its effects on tert-butyl hydroperoxide resistance, we examined resistance to hydrogen peroxide in daf-7 mutants. Assigning control of cellular defenses to dedicated sensory circuits may represent a general cellular-coordination tactic used by animals to regulate induction of self-defenses for hydrogen peroxide and perhaps other threats. We show that the two ASI amphid sensory neurons use a multistep signal relay to control the extent to which target tissues protect C. elegans from hydrogen peroxide.

Figures:

Figure 1: Sensory neurons regulate peroxide resistance inC.elegans.

Figure 2: Sensory neurons regulate peroxide resistance inC.elegans.

Figure 3: ASI sensory neurons secrete DAF-7/TGF to specifically lower the nematodes peroxide resistance.

Figure 4: DAF-7 regulates peroxide resistance robustly.

Figure 5: The DAF-1/TGF receptor functions redundantly in interneurons to regulate peroxide resistance in response to DAF-7/TGF from ASI.

Figure 6: The DAF-1/TGF receptor functions redundantly in interneurons to regulate peroxide resistance in response to DAF-7/TGF from ASI.

Figure 7: DAF-1/TGF-receptor signaling regulates peroxide resistance separately from its role in dauer formation, fat storage, and germline growth.

Figure 8: DAF-1/TGF-receptor signaling regulates peroxide resistance separately from its role in fat storage.

Figure 9: ASI regulates the nematodes peroxide resistance via a TGF-Insulin/IGF1 hormone relay.

Figure 10: ASI regulates the nematodes peroxide resistance via a TGF-Insulin/IGF1 hormone relay.

Figure 11: Reduced DAF-7/TGF signaling upregulates expression of DAF-16/FOXO andSKN-1/NRF target genes.

Figure 12: Reduced DAF-7/TGF signaling upregulates expression of DAF-16/FOXO and SKN-1/NRF target genes.

Figure 13: Food ingestion regulates the nematodes peroxide resistance via DAF-3/coSMAD.

Figure 14: DAF-3/coSMAD increases the nematodes peroxide resistance in response to reduced feeding.

Figure 15: DAF-7/TGF signals that hydrogen-peroxide protection will be provided by catalases fromE. coliand not by catalases fromC. elegans.

Figure 16: DAF-7/TGF signals that hydrogen-peroxide protection will be provided by catalases fromE. coliand not by catalases fromC. elegans.

Mentioned biomedical entities:

Genes: BAG, CTL-1, DAF-1, DAF-16, DAF-2, DAF-3, DAF-5, DAF-7, FLP, FOXO, HLH-30, IL2, KatE, P40, SKN-1, TFEB, ctl-1, ctl-2, ctl-3, daf-16, glr-1, mgl

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, oocyte

Organisms: C. elegans, E. coli, E. faecium, human, mice

Tissues: anterior, egg, eggshell, hypodermis, lens, nervous system

Pathways: N/A

Query Result 11/25

FLP-18 Functions through the G-Protein-Coupled Receptors NPR-1 and NPR-4 to Modulate Reversal Length in Caenorhabditis elegans

Bhardwaj, Ashwani; Thapliyal, Saurabh; Dahiya, Yogesh; Babu, Kavita

The Journal of Neuroscience

Year: 2018 PMID: 29712787 doi: 10.1523/JNEUROSCI.1955-17.2018Citations: 25 Score: 11.69

Keywords hits: ASE: 102, che-1: 0, lsy-6: 0

Abstract:

Animal behavior is critically dependent on the activity of neuropeptides. Reversals, one of the most conspicuous behaviors in Caenorhabditis elegans, plays an important role in determining the navigation strategy of the animal. Our experiments on hermaphrodite C. elegans show the involvement of a neuropeptide FLP-18 in modulating reversal length in these hermaphrodites. We show that FLP-18 controls the reversal length by regulating the activity of AVA interneurons through the G-protein-coupled neuropeptide receptors, NPR-4 and NPR-1. We go on to show that the site of action of these receptors is the AVA interneuron for NPR-4 and the ASE sensory neurons for NPR-1. We further show that mutants in the neuropeptide, flp-18, and its receptors show increased reversal lengths. Consistent with the behavioral data, calcium levels in the AVA neuron of freely reversing C. elegans were significantly higher and persisted for longer durations in flp-18, npr-1, npr-4, and npr-1 npr-4 genetic backgrounds compared with wild-type control animals. Finally, we show that increasing FLP-18 levels through genetic and physiological manipulations causes shorter reversal lengths. Together, our analysis suggests that the FLP-18/NPR-1/NPR-4 signaling is a pivotal point in the regulation of reversal length under varied genetic and environmental conditions.

Summary:

We expressed NPR-4 under the rig-3 promoter using Prig-3:: NPR-4:: sl2:: wrm Scarlet, which is expressed in the AVA command interneuron, and found that this promoter could again rescue the npr-4 mutant phenotype in the npr-1 npr-4 double mutants. This rescue confirmed that NPR-1 could function in sensory neurons and further indicated that NPR-1 could be modulating reversal length through the ASE sensory neurons, which are the only neurons where NPR-1 is expressed under both gpa-3 and flp-5 promoters. These results show that NPR-1 expression in the ASE sensory neurons and NPR-4 expression in the AVA command interneuron are able to rescue the increased activity of the AVA neuron, seen in the npr-1 npr-4 mutant C. elegans. Having found FLP-18 to be required for controlling reverse body bends, we reasoned that changes in the cellular levels of FLP-18 could change the length of reversals, and hence food search strategy in response to different environmental conditions. We found a significant increase in the FLP-18 expression in creb1/crh-1 mutant animals compared with WT control animals, where neurons expressing FLP-18 showed increased expression in creb1/crh-1 mutants. Together, these experiments show that the FLP-18 pathway functioning through NPR-1 and NPR-4 is required for reducing the reversal length during starvation in C. elegans. One discrepancy that we found in these experiments was that there was a significant reduction in reversal length in flp-18 mutants compared with our previous reversal data.

Figures:

Figure 1: Mutants in flp-18 show increased body bends per reversal.

Figure 2: Mutants in flp-18 and its receptors npr-1 and npr-4 show increased AVA activity.

Figure 3: CREB1/CRH-1 regulates FLP-18 expression.

Figure 4: Starvation causes increased FLP-18 levels in C. elegans .

Figure 5: Schematic model for the neuropeptide FLP-18 based modulation of the reversal circuit at the level of sensory neurons and interneurons.

Mentioned biomedical entities:

Genes: B, ChR2, FLP-18, NPR-11, NPR-4, NPR-5, P40, PHB, RIG, UNC-2, crh-1, flp-18, flp-5, gcy-5, sra-6, unc-30, unc-4

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, interneuron, neuron

Organisms: C. elegans

Tissues: dorsal

Pathways: N/A

Query Result 12/25

Antagonistic regulation of salt and sugar chemotaxis plasticity by a single chemosensory neuron in Caenorhabditis elegans

Tomioka, Masahiro; Umemura, Yusuke; Ueoka, Yutaro; Chin, Risshun; Katae, Keita; Uchiyama, Chihiro; Ike, Yasuaki; Iino, Yuichi; Hart, Anne C.; Hart, Anne C.; Copenhaver, Gregory P.; Hart, Anne C.; Copenhaver, Gregory P.; Hart, Anne C.; Copenhaver, Gregory P.; Hart, Anne C.; Copenhaver, Gregory P.

PLOS Genetics

Year: 2023 PMID: 37669262 doi: 10.1371/journal.pgen.1010637.r002 Citations: 4 Score: 11.31

Keywords hits: ASE: 225, che-1: 0, lsy-6: 0

Abstract:

The nematode Caenorhabditis elegans memorizes various external chemicals, such as ions and odorants, during feeding. Here we find that C. elegans is attracted to the monosaccharides glucose and fructose after exposure to these monosaccharides in the presence of food; however, it avoids them without conditioning. The attraction to glucose requires a gustatory neuron called ASEL. ASEL activity increases when glucose concentration decreases. Optogenetic ASEL stimulation promotes forward movements; however, after glucose conditioning, it promotes turning, suggesting that after glucose conditioning, the behavioral output of ASEL activation switches toward glucose. We previously reported that chemotaxis toward sodium ion (Na+), which is sensed by ASEL, increases after Na+ conditioning in the presence of food. Interestingly, glucose conditioning decreases Na+ chemotaxis, and conversely, Na+ conditioning decreases glucose chemotaxis, suggesting the reciprocal inhibition of learned chemotaxis to distinct chemicals. The activation of PKC-1, an nPKC / ortholog, in ASEL promotes glucose chemotaxis and decreases Na+ chemotaxis after glucose conditioning. Furthermore, genetic screening identified ENSA-1, an ortholog of the protein phosphatase inhibitor ARPP-16/19, which functions in parallel with PKC-1 in glucose-induced chemotactic learning toward distinct chemicals. These findings suggest that kinasephosphatase signaling regulates the balance between learned behaviors based on glucose conditioning in ASEL, which might contribute to migration toward chemical compositions where the animals were previously fed.

Summary:

The ASEL neuron responds to glucose and is required for glucose attraction after glucose conditioning. Genetic ablation of ASEL using caspase caused a significant defect in glucose attraction after glucose conditioning, suggesting that the ASEL neuron is required for glucose attraction after glucose conditioning. These changes in pirouette responses to glucose led to avoidance or attraction of glucose in worms without conditioning or those after glucose conditioning, respectively. These behavioral phenotypes caused by pkc-1 expression are consistent with the hypothesis that PKC-1 activity in ASEL promotes both glucose attraction and Na+ avoidance after high-glucose conditioning and the previous report suggesting that PKC-1 activity in ASER promotes Na Cl attraction after high-Na Cl conditioning . Although the pkc-1 mutation promoted Na+ attraction in worms without conditioning, glucose conditioning significantly decreased Na+ chemotaxis. Expression of ensa-1 cDNA only in ASEL significantly rescued both glucose attraction and Na+ avoidance after glucose conditioning, suggesting that ensa-1 functions in ASEL to promote glucose attraction and Na+ avoidance after glucose conditioning. To examine the genetic interaction between ensa-1 and pkc-1, we tested Na+ chemotaxis of the pkc-1 ensa-1 double mutant under naive conditions and after conditioning with Na Cl or glucose.

Figures:

Figure 1: C.elegansmemorizes the presence of monosaccharides during feeding.

Figure 2: The ASEL neuron functions in glucose attraction after glucose conditioning.

Figure 3: Behavioral outputs after glucose concentration changes are switched by glucose conditioning.

Figure 4: Conditioning with glucose and sodium ions decreases chemotaxis to sodium ions and glucose, respectively.

Figure 5: PKC-1 in ASEL promotes glucose attraction and Na+avoidance after glucose conditioning.

Figure 6: ENSA-1 in ASEL promotes glucose attraction and Na+avoidance after glucose conditioning.

Figure 7: A summary of chemotaxis plasticity induced by sugar and salt conditioning.

Mentioned biomedical entities:

Genes: ASER, CFP, Cas9, ChR2, MAST3, PKC-1, PP2A, U6, ensa-1, gcy-5, rgef-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: C. elegans, E. coli, Escherichia coli

Tissues: nervous system

Pathways: N/A

Query Result 13/25

Mechanism of life-long maintenance of neuron identity despite molecular fluctuations

Traets, Joleen JH; van der Burght, Servaas N; Rademakers, Suzanne; Jansen, Gert; van Zon, Jeroen S; Hauf, Silke; Walczak, Aleksandra M; Hauf, Silke; Becskei, Attila; Hauf, Silke

eLife

Year: 2021 PMID: 34908528 doi: 10.7554/eLife.66955.sa0 Citations: 0 Score: 11.15

Keywords hits: ASE: 357, che-1: 1006, lsy-6: 0

Abstract:

Cell fate is maintained over long timescales, yet molecular fluctuations can lead to spontaneous loss of this differentiated state. Our simulations identified a possible mechanism that explains life-long maintenance of ASE neuron fate in Caenorhabditis elegans by the terminal selector transcription factor CHE-1. Here, fluctuations in CHE-1 level are buffered by the reservoir of CHE-1 bound at its target promoters, which ensures continued che-1 expression by preferentially binding the che-1 promoter. We provide experimental evidence for this mechanism by showing that che-1 expression was resilient to induced transient CHE-1 depletion, while both expression of CHE-1 targets and ASE function were lost. We identified a 130 bp che-1 promoter fragment responsible for this resilience, with deletion of a homeodomain binding site in this fragment causing stochastic loss of ASE identity long after its determination. Because network architectures that support this mechanism are highly conserved in cell differentiation, it may explain stable cell fate maintenance in many systems.

Summary:

Therefore, we determined copy numbers and lifetimes of che-1 mRNA and protein. First, we measured the absolute number of che-1 mRNA and protein molecules in ASE neurons. After 24 hr without auxin, gcy-22 mRNA levels had increased significantly, consistent with the recovery of CHE-1:: GFP:: AID levels and chemotaxis to Na Cl in these animals. However, che-1:: GFP:: AID mRNA was absent in p:: che-1:: GFP:: AID embryos at later stages, indicating that this 130 bp che-1 promoter region is not required for initiation of che-1 expression but is important for maintenance of che-1 expression and ASE fate. We show that this resilience of che-1 expression to CHE-1 depletion could be conferred onto other target genes by introducing a 130 bp fragment of the che-1 promoter that surrounds the ASE motif bound by CHE-1, but not the ASE motif itself. A definitive test of the target reservoir buffering mechanism would require direct measurements of CHE-1 binding to the promoter of che-1 and its other targets. If this mechanism is indeed responsible for stable ASE fate maintenance, our work suggests that homeodomain transcription factors are involved, as deleting a HD-TF binding site close to the che-1 ASE motif and specific to the che-1 promoter, caused spontaneous transitions to the OFF state, and loss of cell identity, long after ASE type determination.

Figures:

Figure 1: Loss of ASE neuron fate upon transient CHE-1 depletion.

Figure 2: CHE-1 recovery after depletion.

Figure 3: CHE-1 depletion and NaCl chemotaxis.

Figure 4: che-1mRNA and protein copy numbers in ASE neurons.

Figure 5: CHE-1 copy numbers in larvae and embryos.

Figure 6: che-1mRNA and protein lifetimes.

Figure 7: Stable ON state by preferential binding of CHE-1 to its own promoter.

Figure 8: Dependence of ON state stability on number of targets and cooperativity.

Figure 9: Maintenance ofche-1expression during transient CHE-1 depletion.

Figure 10: Region flanking CHE-1-binding site ensures resilience to CHE-1 depletion.

Figure 11: Controls for promoter region mutants.

Figure 12: AnOtx-related homeodomain transcription factor binding site involved in long-term maintenance of ASE cell fate.

Figure 13: Role of HD-TF-binding site in maintaining CHE-1::GFP expression.

Figure 14: Models of HD-TF action.

Figure 15: NA

Mentioned biomedical entities:

Genes: ADF, ALR-1, ASER, CEH-36, CHE-1, EGL-43, HD, L1, NHR-67, P40, ceh-23, cog-1, g51, gcy-22, histone, lim-6

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: C. elegans, E. coli, humans, mice

Tissues: nervous system, tissue

Pathways: N/A

Query Result 14/25

The tubulin repertoire of Caenorhabditis elegans sensory neurons and its contextdependent role in process outgrowth

Lockhead, Dean; Schwarz, Erich M.; OHagan, Robert; Bellotti, Sebastian; Krieg, Michael; Barr, Maureen M.; Dunn, Alexander R.; Sternberg, Paul W.; Goodman, Miriam B.; Strome, Susan

Molecular Biology of the Cell

Year: 2016 PMID: 27654945 doi: 10.1091/mbc.E16-06-0473 Citations: 38 Score: 11.13

Keywords hits: ASE: 53, che-1: 0, lsy-6: 0

Abstract:

Microtubules contribute to key cellular processes and are composed of -tubulin heterodimers. Neurons in Caenorhabditis elegans express cell typespecific isoforms in addition to a shared repertoire and rely on tubulins for neurite outgrowth. Isoform function varies between in vivo and in vitro contexts.

Summary:

We analyze the effect of null mutations in a subset of these isotypes in vivo and in vitro and demonstrate that mec7 is required for the wild-type outgrowth of TRN processes in vivo and in vitro, whereas mec12 plays a partially redundant role with tba-1 in vivo and is required for wild-type outgrowth in in vitro. To study the effects of tubulin isoforms on TRN process outgrowth, we first sought to understand the contribution of microtubules to the outgrowth of TRNs in vivo. Taken together with prior work establishing that colchicine-treated animals lack 15protofilament microtubules in both ALM and PLM, these results show that disrupting MTs has surprisingly mild effects on the extension of TRN processes in vivo. Having established that microtubules contribute to the outgrowth of the posterior TRNs in vivo, we sought to identify the tubulin genes that are expressed in these TRNs. We detected expression of 5086 proteincoding genes in PLM neurons and similar numbers in AFD and ASER neurons isolated from animals late in larval development or in early adulthood. Similarly, tba1 was the most highly expressed tubulin in AFD and ASER, as well as in PLM, after mec12. We found expression of tbb4 in both AFD and ASER but not in PLM. This finding suggests that process length is tightly controlled in vivo but not in vitro. Although TRN shapes were similar in vitro to those found in vivo, where the ALM neurons are unipolar and the PLM neurons are bipolar, we sought to determine how in vitro morphology was connected to the cell fates of ALM and PLM.

Figures:

Figure 1: Pharmacological disruption of microtubules disrupts PLM length in vivo.

Figure 2: Tubulin isoform expression in PLM.

Figure 3: Mutation of mec7 or combination of mec12 with tba1 or tba2 mutations increases the receptive field gap.

Figure 4: The tba1, tba2, tbb1 , or tbb2 loss-of-function mutants have wild-type touch sensitivity.

Figure 5: TRNs adopt similar morphologies in vitro as they do in vivo.

Figure 6: TRNs primarily adopt ALMlike cell fate and extend processes by a pmk3 independent mechanism in vitro.

Figure 7: mec7 and mec12 are necessary for wildtype process outgrowth in vitro.

Mentioned biomedical entities:

Genes: ASER, Cas9, EGL-5, MEC-12, MEC-7, MTs, PAM, PML, SEG, TBA-1, TBA-2, TBB-1, TBB-2, pBX, tbb-2, ttx-1, tubulin

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, multicellular, neuron

Organisms: C. elegans, human, humans

Tissues: anterior, egg

Pathways: N/A

Query Result 15/25

An Excitatory/Inhibitory Switch From Asymmetric Sensory Neurons Defines Postsynaptic Tuning for a Rapid Response to NaCl in Caenorhabditis elegans

Kuramochi, Masahiro; Doi, Motomichi

Frontiers in Molecular Neuroscience

Year: 2019 PMID: 30687001 doi: 10.3389/fnmol.2018.00484 Citations: 6 Score: 10.97

Keywords hits: ASE: 203, che-1: 9, lsy-6: 0

Abstract:
The neural networks that regulate animal behaviors are encoded in terms of neuronal excitation and inhibition at the synapse. However, how the temporal activity of neural circuits is dynamically and precisely characterized by each signaling interaction via excitatory or inhibitory synapses, and how both synaptic patterns are organized to achieve fine regulation of circuit activities is unclear. Here, we show that in Caenorhabditis elegans, the excitatory/inhibitory switch from asymmetric sensory neurons (ASEL/R) following changes in NaCl concentration is required for a rapid and fine response in postsynaptic interneurons (AIBs). We found that glutamate released by the ASEL neuron inhibits AIBs via a glutamate-gated chloride channel localized at the distal region of AIB neurites. Conversely, glutamate released by the ASER neuron activates AIBs via an AMPA-type ionotropic receptor and a G-protein-coupled metabotropic glutamate receptor. Interestingly, these excitatory receptors are mainly distributed at the proximal regions of the neurite. Our results suggest that these convergent synaptic patterns can tune and regulate the proper behavioral response to environmental changes in NaCl.

Summary:
Therefore, together with our findings in the che-1 mutant, these results suggest that synaptic transmission from ASE neurons is required to regulate AIB activity when Na Cl concentrations change. We further examined how the two distinct ASEL and ASER neuronal responses to changes in Na Cl concentration cooperatively modulate AIB activity. In worms in which transmission from the ASER was rescued by cell-specific expression of UNC-13 in the ASER neuron, we observed that the calcium response in AIB neurons increased just after the Na Cl concentration is decreased. AIB neurons were rapidly inactivated during an upstep in Na Cl concentration, and this inactivation was also observed when the synaptic transmission from the ASEL neuron was specifically rescued upon expressing UNC-13 in the ASEL neuron of unc-13 mutants. These results suggest that the AMPA-type glutamate receptor GLR-1 on the AIB predominantly localize on the proximal neurite to form synapses with the ASER and can receive excitatory glutamate signals from it. Contrary to the ASER-to-AIB excitatory signal, our calcium imaging data showed that ASEL glutamatergic signaling inhibits AIB activity. These results suggest that a few but clear synapses are formed between the ASEL and AIB, and at those synapses, the glutamate-gated chloride ion channel GLC-3 on the AIB receives inhibitory glutamate signals from the ASEL. In this study, we show that the excitatory/inhibitory switch from asymmetric sensory neurons defines smooth and fine transitions of responses to changes in salt concentration in C. elegans postsynaptic neurons.

Figures:
Figure 1: Calcium dynamics in ASEL, ASER and AIB neurons in response to changes in NaCl concentration.

Figure 2: ASEL inhibits AIB whereas ASER stimulates AIB.

Figure 3: Glutamate released by ASE neurons and its receptors on AIB neurons elicit AIB excitation and inhibition.

Figure 4: Localization patterns of the AMPA-type glutamate receptor and the glutamate-gated chloride ion channel on AIB neurites.

Figure 5: The localization patterns of the presynaptic protein RAB-3 in ASEL/R neurons and the postsynaptic AMPA-type glutamate receptors in AIB neurons.

Figure 6: The localization patterns of the presynaptic protein RAB-3 in ASEL/R neurons and the postsynaptic glutamate-gated chloride ion channel in AIB neurons.

Mentioned biomedical entities:
Genes: ASER, GLC-3, RAB-3, UNC-13, avr-14, gcy-5, glr-1, mgl, mgl-2, npr-9

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: C. elegans, Clostridium tetani

Tissues: dorsal, lens

Pathways: N/A

Query Result 16/25

LIN-42, the Caenorhabditis elegans PERIOD homolog, Negatively Regulates MicroRNA Transcription

Perales, Roberto; King, Dana M.; Aguirre-Chen, Cristina; Hammell, Christopher M.; Ruvkun, Gary

PLoS Genetics

Year: 2014 PMID: 25032706 doi: 10.1371/journal.pgen.1004486 Citations: 29 Score: 10.78

Keywords hits: ASE: 42, che-1: 0, lsy-6: 21

Abstract:

During C. elegans development, microRNAs (miRNAs) function as molecular switches that define temporal gene expression and cell lineage patterns in a dosage-dependent manner. It is critical, therefore, that the expression of miRNAs be tightly regulated so that target mRNA expression is properly controlled. The molecular mechanisms that function to optimize or control miRNA levels during development are unknown. Here we find that mutations in lin-42, the C. elegans homolog of the circadian-related period gene, suppress multiple dosage-dependent miRNA phenotypes including those involved in developmental timing and neuronal cell fate determination. Analysis of mature miRNA levels in lin-42 mutants indicates that lin-42 functions to attenuate miRNA expression. Through the analysis of transcriptional reporters, we show that the upstream cis-acting regulatory regions of several miRNA genes are sufficient to promote highly dynamic transcription that is coupled to the molting cycles of post-embryonic development. Immunoprecipitation of LIN-42 complexes indicates that LIN-42 binds the putative cis-regulatory regions of both non-coding and protein-coding genes and likely plays a role in regulating their transcription. Consistent with this hypothesis, analysis of miRNA transcriptional reporters in lin-42 mutants indicates that lin-42 regulates miRNA transcription. Surprisingly, strong loss-of-function mutations in lin-42 do not abolish the oscillatory expression patterns of lin-4 and let-7 transcription but lead to increased expression of these genes. We propose that lin-42 functions to negatively regulate the transcriptional output of multiple miRNAs and mRNAs and therefore coordinates the expression levels of genes that dictate temporal cell fate with other regulatory programs that promote rhythmic gene expression.

Summary:

In many of these scenarios, the expression of distinct miRNAs is tightly controlled and/or the individual steps of miRNA biogenesis are actively regulated at either the transcriptional or post-transcriptional level by sequence-specific transcription factors or RNA-binding proteins, respectively. Importantly, many of the proteins that regulate miRNA biogenesis are highly conserved and mutations in these genes result in a variety of developmental disorders and diseases. The C. elegans heterochronic pathway has been instrumental to our understanding of the principles of miRNA-mediated gene regulation and for the identification of components that are required to control miRNA expression, metabolism and activity. In addition, mutations that alter heterochronic miRNA expression often display strong temporal patterning and behavioral phenotypes, . While the regulatory strategies that dictate patterns of cell fate specification have rapidly emerged through the identification of conserved heterochronic genes, we still lack a deep understanding of how the temporal expression of heterochronic genes are coordinated with aspects of growth and behavior. As a product of this approach, we identified mutations in lin-42 that suppress multiple stage-specific lineage defects associated with heterochronic miRNAs. Analysis of miRNA expression in lin-42 mutant animals suggests that LIN-42 broadly functions to negatively regulate miRNA expression is therefore is likely to act in a variety of pathways that require miRNAs for proper cell fate specification.

Figures:

Figure 1: Mutations inlin-42suppress defects in temporal gene expression and cell lineage phenotypes of heterochronic miRNA mutants.

Figure 2: , bc toggle=yeslin-42 mutations lead to the overexpression of several miRNAs.

Figure 3: , bc toggle=yeslin-42 suppresses neuronal phenotypes associated withlsy-6miRNA-mediated cell fate specification.

Figure 4: The promoters of three different miRNAs display dynamic expression patterns that are coupled to the larval molting cycle.

Figure 5: LIN-42 binds to the putative regulatory regions of both miRNAs and mRNAs.

Figure 6: A model forlin-42function in regulating post-embryonic miRNA expression.

Mentioned biomedical entities:

Genes: Argonaute, Dicer, Drosha, L1, LIN-42, Pasha, lin-4, lin-42

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: C. elegans, human

Tissues: seed

Pathways: N/A

Query Result 17/25

Molecular topography of an entire nervous system

Taylor, Seth R.; Santpere, Gabriel; Weinreb, Alexis; Barrett, Alec; Reilly, Molly B.; Xu, Chuan; Varol, Erdem; Oikonomou, Panos; Glenwinkel, Lori; McWhirter, Rebecca; Poff, Abigail; Basavaraju, Manasa; Rafi, Ibnul; Yemini, Eviatar; Cook, Steven J.; Abrams, Alexander; Vidal, Berta; Cros, Cyril; Tavazoie, Saeed; Sestan, Nenad; Hammarlund, Marc; Hobert, Oliver; Miller, David M.

Cell

Year: 2021 PMID: 34237253 doi: 10.1016/j.cell.2021.06.023 Citations: 334 Score: 10.71

Keywords hits: ASE: 0, che-1: 0, lsy-6: 0

Abstract:

We have produced expression profiles of all 302 neurons of the C. elegans nervous system that match the single cell resolution of its anatomy and wiring diagram. Our results suggest that individual neuron classes can be solely identified by combinatorial expression of specific gene families. For example, each neuron class expresses distinct codes of 23 neuropeptide genes and 36 neuropeptide receptors, delineating a complex and expansive wireless signaling network. To demonstrate the utility of this comprehensive gene expression catalog, we used computational approaches to (1) identify cis-regulatory elements for neuron-specific gene expression and (2) reveal adhesion proteins with potential roles in process placement and synaptic specificity. Our expression data are available at cengen.org and can be interrogated at the web application CengenApp. We expect that this neuron-specific directory of gene expression will spur investigations of underlying mechanisms that define anatomy, connectivity and function throughout the C. elegans nervous system.

Summary:

N/A

Figures:

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 18/25

Genome-wide RNAi screen for regulators of UPRmt in Caenorhabditis elegans mutants with defects in mitochondrial fusion

Haeussler, Simon; Yeroslaviz, Assa; Rolland, Stphane G; Luehr, Sebastian; Lambie, Eric J; Conradt, Barbara

G3: GenesGenomesGenetics

Year: 2021   PMID: 33784383   doi: 10.1093/g3journal/jkab095   Citations: 8   Score: 10.55

Keywords hits: ASE: 96, che-1: 1, lsy-6: 0

Abstract:
Mitochondrial dynamics plays an important role in mitochondrial quality control and the adaptation of metabolic activity in response to environmental changes. The disruption of mitochondrial dynamics has detrimental consequences for mitochondrial and cellular homeostasis and leads to the activation of the mitochondrial unfolded protein response (UPRmt), a quality control mechanism that adjusts cellular metabolism and restores homeostasis. To identify genes involved in the induction of UPRmt in response to a block in mitochondrial fusion, we performed a genome-wide RNAi screen in Caenorhabditis elegans mutants lacking the gene fzo-1, which encodes the ortholog of mammalian Mitofusin, and identified 299 suppressors and 86 enhancers. Approximately 90% of these 385 genes are conserved in humans, and one-third of the conserved genes have been implicated in human disease. Furthermore, many have roles in developmental processes, which suggests that mitochondrial function and their response to stress are defined during development and maintained throughout life. Our dataset primarily contains mitochondrial enhancers and non-mitochondrial suppressors of UPRmt, indicating that the maintenance of mitochondrial homeostasis has evolved as a critical cellular function, which, when disrupted, can be compensated for by many different cellular processes. Analysis of the subsets non-mitochondrial enhancers and mitochondrial suppressors suggests that organellar contact sites, especially between the ER and mitochondria, are of importance for mitochondrial homeostasis. In addition, we identified several genes involved in IP3 signaling that modulate UPRmt in fzo-1 mutants and found a potential link between pre-mRNA splicing and UPRmt activation.

Summary:
All except three genes induce Phsp-6mtHSP70gfp expression when knocked-down in wild-type animals, suggesting that the induction of UPRmt by depletion of these candidates is independent of the loss of fzo-1 Candidates that encode mitochondrial proteins and that induce UPRmt in a wild-type background upon knock-down were included in a recent publication, which reported the systematic identification of mitochondrial inducers of UPRmt . Among the 299 suppressors, only 25 have previously been found to suppress UPRmt induced by other means upon knock-down. These TFs may be specific to UPRmt in fzo-1 but some may generally be involved in UPRmt signaling. In order to determine whether any of the suppressors or enhancers that we identified have previously been shown to interact with fzo-1MFN1,2 or its mammalian orthologs MFN1 or MFN2, we built a gene network containing all known interactions of fzo-1MFN1,2 and its mammalian orthologs MFN1 and MFN2. These could either have been missed in the RNAi screen, be essential in the fzo-1 background or not have a function in mitochondrial homeostasis and, hence, UPRmt signaling. Similar to the approach described above, we used the 16 C. elegans genes currently associated with the GO-term mitochondrial unfolded protein response, identified their human orthologs and included known physical and genetic interactors from the interaction databases BioGRID3.5, Int Act, and mentha to calculate an interaction network containing 2603 genes and 4655 interactions. In this UPRmtome, we identified 129 genes, 36 of which are enhancers and 77 of which are suppressors of fzo-1-induced UPRmt, with a total of 213 interactions. In summary, we identified several genes in our dataset using gene network analysis that have previously not been identified to play a role in UPRmt signaling in C. elegans.

Figures:

Figure 1: Overview of genome-wide RNAi screen for suppressors and enhancers of fzo-1(tm1133) -induced UPR mt .

Figure 2: GO enrichment analysis of suppressors and enhancers of fzo-1(tm1133) -induced UPR mt using DAVID.

Figure 3: Enrichment analysis of transcription factors binding to promotors of candidate genes that suppress or enhance fzo-1(tm1133) -induced UPR mt .

Figure 4: Analysis of a gene networkthe UPR mt ome.

Figure 5: Candidate genes with roles in IP 3 signaling.

Figure 6: , bc toggle=yesmiga-1(tm3621) mutants induce UPR mt and have altered mitochondrial morphology.

Mentioned biomedical entities:

Genes: ARCN1, CTNNB1, ER, FZO-1, HLH-29, HNF4A, HUB1, MAVS, MFN1, MFN2, Mfn1, Mfn2, P40, PI4K, RET2, Ras, SLC25A38, TOR

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, cytosol, photoreceptor cell

Organisms: C. elegans, Human, human, humans, mice

Tissues: seed, tissue

Pathways: N/A

Query Result 19/25

The C. elegans cGMP-Dependent Protein Kinase EGL-4 Regulates Nociceptive Behavioral Sensitivity

Krzyzanowski, Michelle C.; Brueggemann, Chantal; Ezak, Meredith J.; Wood, Jordan F.; Michaels, Kerry L.; Jackson, Christopher A.; Juang, Bi-Tzen; Collins, Kimberly D.; Yu, Michael C.; LEtoile, Noelle D.; Ferkey, Denise M.; Goodman, Miriam B.

PLoS Genetics

Year: 2013 PMID: 23874221 doi: 10.1371/journal.pgen.1003619 Citations: 22 Score: 10.47

Keywords hits: ASE: 71, che-1: 0, lsy-6: 0

Abstract:

Signaling levels within sensory neurons must be tightly regulated to allow cells to integrate information from multiple signaling inputs and to respond to new stimuli. Herein we report a new role for the cGMP-dependent protein kinase EGL-4 in the negative regulation of G protein-coupled nociceptive chemosensory signaling. C. elegans lacking EGL-4 function are hypersensitive in their behavioral response to low concentrations of the bitter tastant quinine and exhibit an elevated calcium flux in the ASH sensory neurons in response to quinine. We provide the first direct evidence for cGMP/PKG function in ASH and propose that ODR-1, GCY-27, GCY-33 and GCY-34 act in a non-cell-autonomous manner to provide cGMP for EGL-4 function in ASH. Our data suggest that activated EGL-4 dampens quinine sensitivity via phosphorylation and activation of the regulator of G protein signaling (RGS) proteins RGS-2 and RGS-3, which in turn downregulate G signaling and behavioral sensitivity.

Summary:

Elegans lacking RGS-3 function are defective in their response to a subset of strong sensory stimuli detected by the ASH sensory neurons. As was previously reported for nave AWC neurons,, GFPEGL-4 was distributed throughout ASH in wild-type animals, with expression seen in both the cytoplasm and the nucleus. To ascertain whether the cytoplasmic or nuclear pools of EGL-4 contribute to the regulation of quinine avoidance, we used the osm-10 promoter to express two modified forms of GFPEGL-4 in egl-4 animals. Furthermore, consistent with a role in dampening sensory sensitivity, overexpression of RGS-2 or RGS-3 in the ASHs of wild-type animals decreased quinine response. If EGL-4 functions in the same pathway as RGS-2 and RGS-3 to regulate chemosensory signaling, then egl-4 rgs-2 and egl-4 rgs-3 double mutant animals should display a behavioral phenotype similar to egl-4 animals and the hypersensitivity should not be additive. To determine whether EGL-4 function in either the ASHs or ASKs is sufficient to regulate quinine response, the cell-selective promoters osm-10, srb-6 and srbc-66 were used to restore EGL-4 function in each neuron pair and animals were assayed for response to 1 mM quinine. We now provide the first direct evidence for EGL-4 function in the ASH nociceptive neurons, and show that it acts to negatively regulate G protein-coupled signal transduction and dampen behavioral sensitivity to the bitter tastant quinine. We found that egl-4 animals respond better than wild-type animals to 10 mM quinine and also avoid dilute levels of quinine that wild-type animals do not respond to.

Figures:

Figure 1: , bc toggle=yesC. elegans EGL-4 regulates quinine sensitivity in ASH.

Figure 2: EGL-4 functions in the cytoplasm to regulate calcium signaling.

Figure 3: RGS proteins are targets of EGL-4.

Figure 4: Guanylyl cyclases act upstream of EGL-4.

Figure 5: EGL-4 does not regulate ASH sensitivity in general.

Figure 6: Model for EGL-4 regulation of nociceptive signaling.

Mentioned biomedical entities:

Genes: Akt, Bad, DAF-5, EGL-10, EGL-4, GCY-27, GCY-33, GCY-34, GPA-3, JNK, KIN-29, Notch, ODR-1, OSM-11, PKG, RGS-2, RGS-3, RGS-7, RGS2, RGS21, RGS3, RGS4, osm-10

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: astrocyte, cell, platelet

Organisms: C. elegans, Mice, mice

Tissues: egg

Pathways: N/A

Query Result 20/25

Forkhead transcription factor FKH-8 cooperates with RFX in the direct regulation of sensory cilia in Caenorhabditis elegans

Brocal-Ruiz, Rebeca; Esteve-Serrano, Ainara; Mora-Martnez, Carlos; Franco-Rivadeneira, Maria Luisa; Swoboda, Peter; Tena, Juan J; Vilar, Maral; Flames, Nuria; Portman, Douglas; Sengupta, Piali; Portman, Douglas; Portman, Douglas

eLife

Year: 2023 PMID: 37449480 doi: 10.7554/eLife.89702.sa2 Citations: 8 Score: 10.32

Keywords hits: ASE: 55, che-1: 6, lsy-6: 0

Abstract:

Cilia, either motile or non-motile (a.k.a primary or sensory), are complex evolutionarily conserved eukaryotic structures composed of hundreds of proteins required for their assembly, structure and function that are collectively known as the ciliome. Ciliome gene mutations underlie a group of pleiotropic genetic diseases known as ciliopathies. Proper cilium function requires the tight coregulation of ciliome gene transcription, which is only fragmentarily understood. RFX transcription factors (TF) have an evolutionarily conserved role in the direct activation of ciliome genes both in motile and non-motile cilia cell-types. In vertebrates, FoxJ1 and FoxN4 Forkhead (FKH) TFs work with RFX in the direct activation of ciliome genes, exclusively in motile cilia cell-types. No additional TFs have been described to act together with RFX in primary cilia cell-types in any organism. Here we describe FKH-8, a FKH TF, as a direct regulator of the sensory ciliome genes in Caenorhabditis elegans. FKH-8 is expressed in all ciliated neurons in C. elegans, binds the regulatory regions of ciliome genes, regulates ciliome gene expression, cilium morphology and a wide range of behaviors mediated by sensory ciliated neurons. FKH-8 and DAF-19 (C. elegans RFX) physically interact and synergistically regulate ciliome gene expression. C. elegans FKH-8 function can be replaced by mouse FOXJ1 and FOXN4 but not by other members of other mouse FKH subfamilies. In conclusion, RFX and FKH TF families act jointly as direct regulators of ciliome genes also in sensory ciliated cell types suggesting that this regulatory logic could be an ancient trait predating functional cilia sub-specialization.

Summary:

To avoid the dauer constitutive phenotype of daf-19 null animals, we analyzed reporter expression in daf-19 daf-12 double mutants and daf-12 was used as control, as previously reported. In daf-12 worms, all reporters show broad activity in the ciliated system, with mean reporter expression in at least 30 ciliated neurons, except for mks-1 and osm-5 reporters that showed expression in less than 20 cells, suggesting other enhancers outside the analyzed sequences might drive expression in additional ciliated neurons. Using an independent set of sc-RNAseq data from young adult, we found that among the 10 TF candidates only FKH-8 expression is detected in all 25 different types of sensory ciliated neurons, suggesting it could be a good candidate to act together with DAF-19 in the regulation of ciliome gene expression. Finally, we interrogated 446 published ChIP-seq datasets, corresponding to 259 different TFs, for nearby binding to the ciliome gene list. In addition, there is a high gene expression correlation for the 73 core ciliome genes and daf-19 or fkh-8 expression but not with other TFs. These results, together with the stronger phenotypes of daf-19 mutants, indicate that although both TFs are required for ciliome gene expression, RFX TFs display more instructive functions in ciliome gene expression. Considering that DAF-19 and FKH-8 share several ciliome gene targets, bind the regulatory regions of ciliome genes in close proximity and both can physically interact, we aimed to assess if DAF-19 and FKH-8 cooperate in the regulation of ciliome gene expression. Importantly, we find that conserved functionality is not observed for any vertebrate FKH TFs as expression of mouse FoxI1, a FKH TF involved in the development of several tissues but not reported to control cilia gene expression, does not rescue fkh-8 mutant phenotype. In summary, our results unravel the functional conservation between FKH-8 and specific mouse members of the FKH family, which have already been described to act together with RFX TFs in the regulation of ciliome gene expression in motile cilia cell types. RFX are the only TFs known to be involved in the direct coregulation of ciliome gene expression both in cell types with motile and sensory cilia.

Figures:

Figure 1: FKH-8 is a candidate for direct regulation of ciliome gene expression inC. elegans.

Figure 2: Ciliome reporters used in this work.

Figure 3: Reporter expression for some core ciliome genes is abolished indaf-19(m86) mutant.

Figure 4: Available -omics data identifies FKH-8 as a candidate transcriptional regulator of ciliome genes inC.elegans.

Figure 5: FKH-8 is expressed in sensory ciliated neurons, binds ciliome genes near DAF-19 X-boxes and physically interacts with DAF-19.

Figure 6: fkh-8expression along ciliated system development.

Figure 7: FKH-8 binds near X-BOX motifs.

Figure 8: f.fkh-8(tm292) is a hypomorphic recessive allele.

Figure 9: f. Functional characterization of putative FKH sites in cis-regulatory modules of two core ciliome components.

Figure 10: FKH-8 and DAF-19 exhibit crosstalk and synergistic effects in the transcriptional regulation of ciliome genes.

Figure 11: Lack of FKH-8 has no major effect on DAF-19 expression.

Figure 12: FKH-8 and DAF-19 show synergistic effects in the transcriptional regulation of the ciliome.

Figure 13: fkh-8(vlc43) null mutants display morphological defects in cilia.

Figure 14: fkh-8(vlc43) null mutants display morphological defects in cilia using physical immobilization with polystyrene beads.

Figure 15: FKH-8 is required for the correct display of several sensory mediated behaviors.

Figure 16: FKH-8 is not required for correct display of mechanosensory behaviors mediated by non-ciliated neurons.

Figure 17: Mammalian FKH TFs with known motile cilia regulatory functions can rescuefkh-8mutant phenotype.

Mentioned biomedical entities:

Genes: ADF, B, BAG, Cas9, DAF-19, FHL, FKH, FKH-8, FLP, FOXI1, FOXJ1, FOXN4, FoxI, FoxI1, FoxJ1, FoxN4, Foxj1, M13, N2, P0, PHB, RFX, RFX2, RFX3, osm-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, ciliated cell, cilium, multicellular, neuron

Organisms: C. elegans, Danio rerio, Mus musculus, human, humans, mouse, zebrafish

Tissues: peripheral

Pathways: N/A

Query Result 21/25

Identification of Odor Blend Used by Caenorhabditis elegans for Pathogen Recognition

Worthy, Soleil E; Rojas, German L; Taylor, Charles J; Glater, Elizabeth E

Chemical Senses

Year: 2018 PMID: 29373666 doi: 10.1093/chemse/bjy001 Citations: 21 Score: 10.16

Keywords hits: ASE: 38, che-1: 2, lsy-6: 0

Abstract:

Animals have evolved specialized pathways to detect appropriate food sources and avoid harmful ones. Caenorhabditis elegans can distinguish among the odors of various species of bacteria, its major food source, but little is known about what specific chemical cue or combination of chemical cues C. elegans uses to detect and recognize different microbes. Here, we examine the strong innate attraction of C. elegans for the odor of the pathogenic bacterium, Serratia marcescens. This initial attraction likely facilitates ingestion and infection of the C. elegans host. Using solid-phase microextraction and gas chromatography coupled with mass spectrometry, we identify 5 volatile odors released by S. marcescens and identify those that are attractive to C. elegans. We use genetic methods to show that the amphid chemosensory neuron, AWCON, senses both S. marcescens-released 2-butanone and acetone and drives attraction to S. marcescens. In C. elegans, pairing a single odorant with food deprivation results in a reduced attractive response for that specific odor. We find that pairing the natural odor of S. marcescens with food deprivation results in a reduced attraction for the natural odor of S. marcescens and a similar reduced attraction for the synthetic blend of acetone and 2-butanone. This result indicates that only 2 odorants represent the more complex odor bouquet of S. marcescens. Although bacterial-released volatiles have long been known to be attractive to C. elegans, this study defines for the first time specific volatile cues that represent a particular bacterium to C. elegans.

Summary:

Second, after the odor of S. marcescens is paired with an adverse experience, what volatiles are later used to recognize and avoid S. marcescens? In this study, we define the attractive volatiles necessary for recognition of S. marcescens by C. elegans. These results indicate that the number of AWCON neurons can determine S. marcescens preference: animals with 2 AWCON neurons have an even stronger preference for S. marcescens than wild-type animals with one AWCON neuron. Taken together, C. elegans had a strong preference for the odor of S. marcescens over the odor of E. coli in a bacterial choice assay. Mutants selectively lacking AWCON or both AWC neurons had a reversed preference favoring E. coli, suggesting that in the absence of AWC neurons, non-AWC neurons can drive either attraction toward E. coli or repulsion from S. marcescens odors. Does the AWCON neuron which mediates the olfactory preference for S. marcescens over E. coli also mediate the attraction to acetone and 2-butanone? We odor conditioned worms with different single odorants found in S. marcescens headspace paired with food deprivation for 90 min and then tested the conditioned and nave worms in a S. marcescens and E. coli bacterial choice assay. In an aversive learning assay where the odor of S. marcescens is paired with food deprivation, C. elegans reduces its chemotaxis similarly for the blend of acetone and 2-butanone and the odor of S. marcescens, indicating that the blend of acetone and 2-butanone represents S. marcescens to C. elegans. The standard C. elegans laboratory strain N2 has a strong preference for the odor of S. marcescens over the odor of E. coli HB101 in an olfactory choice assay.

Figures:

Figure 1: , b Caenorhabditis elegans prefers Serratia marcescens over E. coli .

Figure 2: Gas chromatography-mass spectrometry of headspace of Serratia marcescens , E. coli and LB media control.

Figure 3: Major attractive volatiles released by Serratia marcescens are acetone and 2-butanone.

Figure 4: AWC ON mediates Caenorhabditis elegans preference for Serratia marcescens .

Figure 5: AWC ON mediates chemotaxis to acetone and Escherichia coli with added acetone and 2-butanone.

Figure 6: Aversive olfactory conditioning with odor of Serratia marcescens reduces preference to 2-butanone and acetone.

Figure 7: Aversive olfactory conditioning with single odorants reduces preference for Serratia marcescens .

Mentioned biomedical entities:

Genes: N2, P40, ceh-36, lim-4, nsy-1, tax-4

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: C. elegans, E. coli, Escherichia coli, human

Tissues: N/A

Pathways: N/A

Query Result 22/25

A lineage-resolved molecular atlas of C. elegans embryogenesis at single cell resolution

Packer, Jonathan S.; Zhu, Qin; Huynh, Chau; Sivaramakrishnan, Priya; Preston, Elicia; Dueck, Hannah; Stefanik, Derek; Tan, Kai; Trapnell, Cole; Kim, Junhyong; Waterston, Robert H.; Murray, John I.

Science (New York, N.Y.)

Year: 2019 PMID: 31488706 doi: 10.1126/science.aax1971 Citations: 261 Score: 10.03

Keywords hits: ASE: 0, che-1: 0, lsy-6: 0

Abstract:

C. elegans is an animal with few cells, but a striking diversity of cell types. Here, we characterize the molecular basis for their specification by profiling the transcriptomes of 86,024 single embryonic cells. We identify 502 terminal and pre-terminal cell types, mapping most single cell transcriptomes to their exact position in C. elegans invariant lineage. Using these annotations, we find that: 1) the correlation between a cells lineage and its transcriptome increases from mid to late gastrulation, then falls dramatically as cells in the nervous system and pharynx adopt their terminal fates; 2) multilineage priming contributes to the differentiation of sister cells at dozens of lineage branches; and 3) most distinct lineages that produce the same anatomical cell type converge to a homogenous transcriptomic state.

Summary:

N/A

Figures:

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 23/25

Evolution of lateralized gustation in nematodes

NA

eLife

Year: 2025 PMID: 40586702 doi: 10.7554/eLife.103796.3.sa0 Citations: 0 Score: 9.96

Keywords hits: ASE: 177, che-1: 112, lsy-6: 7

Abstract:

Animals with small nervous systems have a limited number of sensory neurons that must encode information from a changing environment. This problem is particularly exacerbated in nematodes that populate a wide variety of distinct ecological niches but only have a few sensory neurons available to encode multiple modalities. How does sensory diversity prevail within this constraint in neuron number? To identify the genetic basis for patterning different nervous systems, we demonstrate that sensory neurons in Pristionchus pacificus respond to various salt sensory cues in a manner that is partially distinct from that of the distantly related nematode Caenorhabditis elegans. Previously we showed that P. pacificus likely lacked bilateral asymmetry (Hong et al., 2019). Here, we show that by visualizing neuronal activity patterns, contrary to previous expectations based on its genome sequence, the salt responses of P. pacificus are encoded in a left/right asymmetric manner in the bilateral ASE neuron pair. Our study illustrates patterns of evolutionary stability and change in the gustatory system of nematodes.

Summary:

Since NH4I sensation is affected by silencing of che-1 neurons but is unaffected in che-1 mutants, ASE function may be more greatly impacted by the silencing of ASE than by the loss of che-1. To assess whether Ppa ASE neurons show the same lateralized response to salt ions as Cel ASE neurons, we generated transgenic P. pacificus lines that express RCaMP in ASE neurons and assessed calcium responses to attractive salts. Specifically, although we observed weaker or comparable responses in AFD neurons when compared to either ASE neurons response toward 250 mM NH4 Cl, Na Cl, and NH4I, the AFD responses were more robust than ASER toward 25 mM NH4 Cl, Na Cl, and NH4I. However, the responses to 250 mM Na Cl were not significantly reduced in either the ASEL or ASER neuron in the gcy-22.3 mutant. We show that three neuron types each have distinct calcium responses to specific ion concentrations, revealing diversity at the single neuron level. RLH, OH, and SHC contributed to the analysis and writing of the work.

Figures:

Figure 1: A comparison of chemotaxis responses to water-soluble ions betweenP. pacificusandC. elegans.

Figure 2: P. pacificus che-1expression in ASE and AFD amphid neurons.

Figure 3: Co-expression ofgcy-22.3p::GFPandttx-1p::RFPin ASER.

Figure 4: che-1expressing amphid neurons are required for sensing water-soluble ions inP. pacificus.

Figure 5: Theche-1p::HisCl1transgene is necessary for histamine-dependent knockdown of salt attraction.

Figure 6: Molecular lesions forP. pacificus che-1andgcy 22.3.

Figure 7: ASEL and ASER responses to different concentrations of NH4Cl, NaCl, and NH4I.

Figure 8: Negative controls of ASE and AFD neuron responses to green light.

Figure 9: Individual traces and heatmaps of ASE responses to various salts.

Figure 10: Microfluidic apparatus used for calcium imaging.

Figure 11: Combined AFD responses in comparison to ASE left or right neuron responses to NH4Cl, NaCl, and NH4I.

Figure 12: Individual traces and heatmaps of AFD responses to various salts.

Figure 13: AFDL/R responses to different concentrations of NH4Cl, NaCl, and NH4I.

Figure 14: The laterally asymmetric expression ofgcy-22.3is dependent on the zinc finger transcription factor CHE-1.

Figure 15: ASE and AFD responses in wildtype compared togcy-22.3mutants.

Figure 16: AFDL and AFDR responses to NH4Cl and NaCl in wildtype andgcy-22.3mutant.

Figure 17: Chemotaxis responses to individual ions ingcy-22.3mutants.

Mentioned biomedical entities:

Genes: ADF, ASER, CHE-1, Cas9, Cel, HisCl1, OTX, Ppa, RIP, SHC, TTX-1, Twist, cog-1, die-1, fozi-1, gcy-5, lim-6, lin-4, lsy-6, rGC, ttx-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: neuron

Organisms: C. elegans, Pristionchus pacificus, human

Tissues: N/A

Pathways: N/A

Query Result 24/25

Sensory plasticity caused by up-down regulation encodes the information of short-term learning and memory

Wang, Ping-Zhou; Ge, Ming-Hai; Su, Pan; Wu, Piao-Ping; Wang, Lei; Zhu, Wei; Li, Rong; Liu, Hui; Wu, Jing-Jing; Xu, Yu; Zhao, Jia-Lu; Li, Si-Jia; Wang, Yan; Chen, Li-Ming; Wu, Tai-Hong; Wu, Zheng-Xing

iScience

Year: 2025   PMID: 40224011   doi: 10.1016/j.isci.2025.112215   Citations: 0   Score: 9.87

Keywords hits: ASE: 277, che-1: 0, lsy-6: 0

Abstract:

Learning and memory are essential for animals well-being and survival. The underlying mechanisms are a major task of neuroscience studies. In this study, we identified a circuit consisting of ASER, RIC, RIS, and AIY, is required for short-term salt chemotaxis learning (SCL) in C.elegans. ASER NaCl-sensation possesses are remodeled by salt/food-deprivation pared conditioning. RIC integrates the sensory information of NaCl and food availability. It excites ASER and inhibits AIY by tyramine/TYRA-2 and octopamine/OCTR-1 signaling pathways, respectively. By the salt conditioning, RIC NaCl calcium response to NaCl is depressed, thus, the RIC excitation of ASER and inhibition of AIY are suppressed. ASER excites RIS by FLP-14/FRPR-10 signaling. RIS inhibits ASER via PDF-2/PDFR-1 signaling in negative feedback. ASER sensory plasticity caused by RIC plasticity and RIS negative feedback are required for both learning and memory recall. Thus, the sensation plasticity encodes the information of the short-term SCL that facilitates animal adaptation to dynamic environments.

Summary:

Thus, we used 50 mM Na Cl in this study. Moreover, ASER sensory plasticity in the salt-conditioned flp-14 and ASER-specific flp-14-rescued animals almost phenocopied that in the frpr-10 mutant and RIS-specific frpr-10 rescued animals. Based on the above results, we conclude that FLP-14/FRPR-10 signaling is not only required for SCL but also for ASER plasticity and that sensory plasticity may be a mechanism of the short-term learning and memory. Given that FRPR-10 signaling in RIS is required for both SCL and ASER plasticity, RIS should regulate ASER sensory plasticity in the process of SCL. The SCL phenotype was restored to the WT only by the genetic reconstitution of both tyra-2 and frpr-10, not by that of a single gene of tyra-2 or frpr-10. Thus, we analyzed the correlation between ASERs sensory plasticity and animals learning index in the tested animals, including the WT N2, the mutant, and transgenic animals of flp-14, ASER: flp-14, frpr-10, RIS: frpr-10, pdf-2, RIS: pdf-2, tdc-1, RIC: tdc-1, tyra-2, ASER: tyra-2, and RIC:: Te Tx. For mock conditioning, the worms were treated with Na Cl-free CTX buffer using the same procedure as that for the salt conditioning. The tested wild-type N2 animals were conditioned by the association of food-deprivation with the treatment of Na Cl-free CTX solution and 100 mM Na Cl-containing CTX solution with varied treatment duration, respectively. A test of chemotaxis to Na Cl or salt chemotaxis learning was performed on a test NGM plate in a Petri dish of 9 cm diameter at room temperature of 20C.

Figures:

Figure 1: Neuropeptidergic FLP-14/FRPR-10 signaling is required for salt chemotaxis learning

Figure 2: The FLP-14/FRPR-10 signaling required for salt chemotaxis learning may transmit neurotransmission from ASER to RIS

Figure 3: FLP-14/FRPR-10 signaling engages in ASER sensory plasticity

Figure 4: PDF-2/PDFR-1 signaling pathway prompts salt chemotaxis learning

Figure 5: RIC drives salt chemotaxis learning via the signaling of tyraminergic TYRA-2 in ASER and octopaminergic OCTR-1 in AIY

Figure 6: ASER sensory plasticity generated by RIC-ASER-RIS circuit encodes the information of salt chemotaxis learning

Figure 7: Time course of the ASER sensory plasticity and working model for the mechanism of ASER plasticity and short-term salt chemotaxis learning

Mentioned biomedical entities:

Genes: ACY-1, ASER, AmpR, B, EGL-3, Gs, HisCl1, IL2, INS-1, K, LGC-55, N2, NPR-11, NPR-2, NPR-4, NPR-5, OCTR-1, P40, S6K, SCL, SER-3, SER-6, TYRA-2, TYRA-3, flp-14, goa-1, gsa-1, pdfr-1, tachykinin, tyra-2, unc-25

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: Escherichia coli

Tissues: lens, seed, tissue

Pathways: N/A

Query Result 25/25

The Bicoid Class Homeodomain Factors ceh-36/OTX and unc-30/PITX Cooperate in C. elegans Embryonic Progenitor Cells to Regulate Robust Development

Walton, Travis; Preston, Elicia; Nair, Gautham; Zacharias, Amanda L.; Raj, Arjun; Murray, John Isaac; Ahringer, Julie

PLoS Genetics

Year: 2015 PMID: 25738873 doi: 10.1371/journal.pgen.1005003 Citations: 25 Score: 9.80

Keywords hits: ASE: 33, che-1: 0, lsy-6: 0

Abstract:

While many transcriptional regulators of pluripotent and terminally differentiated states have been identified, regulation of intermediate progenitor states is less well understood. Previous high throughput cellular resolution expression studies identified dozens of transcription factors with lineage-specific expression patterns in C. elegans embryos that could regulate progenitor identity. In this study we identified a broad embryonic role for the C. elegans OTX transcription factor ceh-36, which was previously shown to be required for the terminal specification of four neurons. ceh-36 is expressed in progenitors of over 30% of embryonic cells, yet is not required for embryonic viability. Quantitative phenotyping by computational analysis of time-lapse movies of ceh-36 mutant embryos identified cell cycle or cell migration defects in over 100 of these cells, but most defects were low-penetrance, suggesting redundancy. Expression of ceh-36 partially overlaps with that of the PITX transcription factor unc-30. unc-30 single mutants are viable but loss of both ceh-36 and unc-30 causes 100% lethality, and double mutants have significantly higher frequencies of cellular developmental defects in the cells where their expression normally overlaps. These factors are also required for robust expression of the downstream developmental regulator mls-2/HMX. This work provides the first example of genetic redundancy between the related yet evolutionarily distant OTX and PITX families of bicoid class homeodomain factors and demonstrates the power of quantitative developmental phenotyping in C. elegans to identify developmental regulators acting in progenitor cells.

Summary:

Computer-aided cell tracking of mutant embryos can confirm these regulators by identifying a wide range of pleiotropic defects, from wholesale fate transformations to subtle defects in cell migration or division timing ,,. Many previous studies of TF function relied on reporter gene expression to infer developmental defects. We conclude that ceh-36 is required for robust larval viability but not for gross morphology or embryonic viability. Multiple pharyngeal gland cell precursors had cell cycle and position defects in ceh-36 mutants. For example, the daughters of the MSaapapa cell normally produce a pharyngeal gland cell and a programmed cell death and the early division of this cell was the largest division-timing defect we observed in ceh-36 mutants. Combinations of reporters with ceh-36 were created using a mating strategy that did not produce heterozygous ceh-36 hermaphrodites at any step or else were verified using PCR. Combinations of unc-30 and ceh-36 were created using nT1q Is51 to balance unc-30 while testing for ceh-36 by PCR. A similar procedure was used to score survival of ceh-36 worms carrying ww Is19. All strains were grown at 20C for over two generations before young adult hermaphrodites were dissected at room temperature in egg buffer and embryos with four or more cells were mounted into a solution of 20m beads in egg buffer/methyl cellulose.

Figures:

Figure 1: CEH-36 protein reporter and endogenous message are expressed dynamically in ABpxp and other lineages.

Figure 2: Examples of cell lineage and position defects observed inceh-36mutants.

Figure 3: Lineage distribution and penetrance of cellular phenotypes inceh-36mutants.

Figure 4: Embryonic coexpression patterns ofC.elegansOTX and PITX factors.

Figure 5: Examples of phenotypes inunc-30;ceh-36double mutants.

Figure 6: unc-30;ceh-36 double mutant lineage phenotypes.

Figure 7: , bc toggle=yesceh-36 andunc-30regulatemls-2and excretory system coalescence.

Figure 8: Complementary hierarchical and lineage accumulation models for developmental regulation.

Figure 9: , btd align=left rowspan=1 colspan=112%

Mentioned biomedical entities:

Genes: CEH-36, CRX, OTX, OTX2, Unc, ceh-36, histone, unc-30

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: C. elegans

Tissues: egg, epithelial tube, tissue

Pathways: N/A